

The Fiscal Consequences of Black Disenfranchisement during Jim Crow for Southern Whites

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Abstract

How did the onset of Jim Crow reshape Southern fiscal priorities? Using a new dataset of annual state spending in fourteen Southern states (1880-1910), we find that although the level of total spending remains unchanged, the composition shifted: redistributive spending on common schools and pensions contracted, while expenditures on higher education, prisons, and debt expanded. Within states, however, state government spending paradoxically increased in counties with larger Black populations, despite the documented collapse in state resources going to Blacks. We trace this shift in spending away from Whites statewide towards Whites in high Black-share counties to skews in representation: population-based apportionment that continued to count disenfranchised Blacks meant that legislative majorities could be secured by winning counties with only 27-40% of each state's White population. Our findings suggest that targeted redistribution to specific low-income Whites was a previously unrecognized contributor to the rise of the Solid South, along with malapportionment.

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1 Introduction

The systematic disenfranchisement of Southern Blacks that precipitated the Jim Crow era is arguably the most significant instance of democratic reversal in American history. Instigated primarily by rural elites in the former highly enslaved “Black Belt” regions across the South, each of the eleven former states of the Confederacy adopted suffrage restrictions, such as poll taxes and literacy tests, between 1890 and 1906 that formally removed the voter eligibility of most Blacks (e.g., Greenberger 2022; Keele et al. 2021; Gray and Jenkins 2024; Perman 2003; Valelly 2004). Scholars have documented the devastating collapse of Black political representation and access to public resources that followed (e.g., Naidu 2012; Margo 2007), along with its long-run consequences for Black economic and social welfare (Althoff and Reichardt 2024; Cook et al. 2023; Derenoncourt et al. 2024). Yet we know less about how this reversal reshaped the fiscal priorities of Southern state governments, or whether the disenfranchisement of Black citizens served the material interests of most Southern Whites. Did it, as some historians suggest (e.g., Woodward 1951), give rise to a “progressivism for whites only”?

In this paper we answer this question using an original annual series of state government expenditures from 1880 to 1910 for fourteen Southern states that relied most heavily on enslaved labor in 1860 and consequently had sizable postwar Black populations rooted in slavery rather than migration.¹ At the state level, we examine changes in the composition of spending across seven major categories—common schools, Confederate pensions, higher education, prisons, debt service, courts, and asylums—which together account for an average of roughly 72% of outlays. As detailed later, some of these items—common schools and pensions—clearly redistributed downward, while others—higher education, prisons, and probably debt service—favored elites, allowing us to trace changes in the distributional orientation of state budgets. At the county level, we analyze the allocation of school funds and pensions to identify which White constituencies gained or lost. We estimate these effects using staggered event-study difference-in-differences designs (DiD) at the state-year and county-year levels.

¹We include every state in which 10% or more of its total population was enslaved in 1860.

We find that total state expenditures do not decline after disenfranchisement, but their composition changes markedly: redistributive spending directed at Whites contracted, while expenditures more aligned with elite interests increased. Within states, however, common-school allocations rose in high-Black-share counties and fell elsewhere, indicating re-targeting rather than general expansion or contraction. Confederate pensions followed a similar pattern—a particularly revealing result because they were entirely state-funded and directed toward lower-income White citizens. These findings are robust across alternative specifications and tests of potential threats to inference. Taken together, the results show that Black disenfranchisement resulted in a general decline in average redistributive spending for Whites statewide in conjunction with an increase in targeted redistribution for Whites in specific geographies.

We then turn to explaining the politics behind this shift in fiscal spending. Building on an established literature (Key 1984; Kousser 1974; Mickey 2015), we trace the mechanism to representation in the state legislatures. Black disenfranchisement led to the overrepresentation of Whites in areas with large Black populations because legislative seats continued to be apportioned according to *total* population—rather than the number of eligible or actual voters—even after Black citizens lost the right to vote. Since disenfranchised Black residents remained in the apportionment base and were highly concentrated geographically, Whites in high-Black-share counties gained disproportionate legislative power. We demonstrate that this distortion led to a sharp decline in what we call the *Pivotal White Share*—the minimal fraction of a state’s White population located in the smallest set of counties capable of commanding a legislative majority. Using an original dataset on county seat allocations in both chambers, we estimate that Democrats could control every Southern state legislature by winning plurality elections in counties containing between 27 and 40% of the statewide White population.² Furthermore, this immediate contraction in the winning coalition is consistent

²Our Pivotal White Share can be read as an empirical counterpart to Caughey’s “selectorate”: whereas Caughey (2018) identifies the selectorate governing Southern members of Congress and statewide officeholders, our measure captures its size and location for each state legislature after Black disenfranchisement. The distinction matters because Caughey (2018, p. 145) finds that Southern politicians—

with the observed re-targeting of redistributive benefits toward Whites in high-Black-share counties. In short, Black disenfranchisement restructured the electoral foundations of Southern politics, narrowing elites' winning coalitions and shifting their fiscal incentives—from broad, statewide redistribution to selective transfers directed at a pivotal subset of Whites.

This paper makes three contributions. First, it provides the first systematic account of how Black disenfranchisement affected overall state spending and its distribution, most of which benefited White Southerners. We do so using the most complete and detailed dataset of Southern state expenditures available for this period. Existing research has documented many dimensions of this episode, including the devastating effects of the formal disenfranchisement of Blacks across the eleven former Confederate states between 1890 and 1906 on Black welfare and access to public resources (e.g., Althoff and Reichardt 2024; Cook et al. 2023; Derenoncourt et al. 2024; Margo 2007). Yet no study has systematically isolated its effects on the composition of state spending, comparing redistributive expenditures with other categories and tracing the consequences for Southern Whites. Our data complement work showing that taxation and public spending in the South were generally lower than elsewhere (Alston and Ferrie 2007; Galor et al. 2009; Jensen et al. 2024), while allowing us to examine variation in redistribution within the region. Our findings also extend Margo (2007) and Kousser (1974, 1980). Margo shows that Black disenfranchisement contributed to an increase in public (state *and* local) education spending on White students relative to their Black counterparts, especially in high-Black-share areas. Kousser documents growing educational disparities among Whites, which he attributes to changes in local taxation. Whereas these studies emphasize how local political power shaped racial and class inequalities in education spending, we show how disenfranchisement altered the within-state distribution of state education funds among Whites through its effects on legislative representation. Our contribution therefore identifies a distinct institutional mechanism and its

especially those in federal office—became increasingly responsive to mass White preferences over the middle third of the twentieth century, with state legislators the exception: “state policies in the region were entirely unresponsive to the preferences of... all white adults... and were indeed biased toward ‘the haves.’”

consequences for inequality in state spending among White Southerners.

Second, our study bridges three literatures in political economy that are typically examined separately: (1) the redistributive consequences of democratic change, (2) the relationship between diversity, group conflict, and public spending, and (3) the effects of malapportionment on distributive politics. While each of these literatures speaks to aspects of our case, key features of our setting fall outside the scope of existing theories in these traditions. For instance, although the fiscal effects of democratic expansions have been central to research on redistribution (Meltzer and Richard 1981; Acemoglu et al. 2015; Cascio and Washington 2014), less work has focused on democratic contractions, particularly in settings of high diversity and intergroup conflict.³ Likewise, the primary models in the literature on diversity and redistribution rarely consider the effects of changes in democracy (e.g., Alesina et al. 1999; Habyarimana et al. 2007). Finally, although the effects of malapportionment on public spending have been modeled and empirically assessed (e.g., Ansolabehere et al. 2002; Ansolabehere and Snyder 2008; Horiuchi and Saito 2003; Ardanaz and Scartascini 2013; Beramendi and Jensen 2019; Beramendi and Rogers 2022), few studies explore how representational bias interacts with suffrage restrictions and ethnic politics. Our work thus offers a clear case of the fiscal consequences of democratic reversal in the presence of institutional differences and social divisions. Our evidence indicates that outcomes depend on how the political marginalization of a given group alters the share of lower-income members of the dominant group required to sustain elite power. Black disenfranchisement reduced the proportion of Whites needed for rural elites to maintain control, thereby diminishing their incentives to redistribute.

Lastly, and perhaps most importantly, we illuminate the political economy of the Solid South—the long period of uncontested Democratic rule that lasted until the 1960s. While scholars have long recognized that this bias to state-legislative representation was central to sustaining this one-party dominance (e.g., Key 1984; Mickey 2015), the fiscal consequences of these institutional arrangements for Southern Whites, and what these indicated about the underpinnings of the Solid South have re-

³See, for instance, Lieberman (2003); Suryanarayan and White (2021) for notable exceptions. We discuss these works below.

mained underexplored. We identify the institutional source of the puzzling disjuncture that Caughey (2018) documents between state and federal representation—apportionment rules that, by continuing to count the disenfranchised, tethered state legislatures to a narrow, rural White electorate—and we uncover a previously overlooked pillar of Democratic hegemony: the geographically targeted redistribution that secured the loyalty of pivotal Whites. Black disenfranchisement, interacting with political geography, apportionment rules, and targeted state spending, thus laid both the institutional and fiscal foundations of the Solid South.

The remainder of the paper proceeds as follows. We first outline the key elements of our case and situate it within the theoretical literatures on distributive spending in political economy. Next, we assess whether Southern Whites benefited fiscally from Black disenfranchisement and analyze why, politically, some low-income Whites gained while the majority did not. We conclude by discussing the broader implications of these findings for the study of redistribution and democratic reversal.

2 Context: The Establishment of the Jim Crow South

The entrenchment of rural elite, one-party rule by the early 20th century was the result of a long-term campaign waged by these rural elites to regain the power they lost as a result of the Civil War, emancipation, and Congressional Reconstruction. Prior to the North's victory in the US Civil War (1861-1865), the economies and political systems of Southern states were dominated by “planters,” a small, rural elite whose wealth was derived from the enslavement of millions of Black agricultural laborers (McCurry 2012; Wright 1978). Defeat not only resulted in the emancipation of more than 40% of the Confederacy's population but also guaranteed the right to vote to the formerly enslaved with the Congressional Reconstruction Acts of 1867 and 1868 (Foner 2014). While the redistributive threat posed by Republican “Reconstruction” governments had been violently eliminated by 1877, adult Black males formally retained the right to vote, continuing to pose a threat to the political dominance of the Democratic Party (Kousser 1974; Perman 2003; Valelly 2004). In an environment of full adult male suffrage, suppressing this electoral challenge required extensive intimidation, violence, electoral manipulation, and fraud (Woodward 1951; Epperly et al. 2020; Jones et al. 2017; Weaver

2019). It also required a degree of progressive taxation and public spending on redistributive goods that, although modest by Northern standards, exceeded the levels previously seen in the South outside of the Reconstruction period (Jensen et al. 2024).⁴

Despite the widespread use of these illegal methods and the implementation of some redistributive fiscal policies, the region remained far from the one-party “Solid South” it would become in the 20th century. Instead, non-Democratic Party candidates for governor and the state legislature continued to receive substantial shares of the vote in many Southern states in the immediate post-Reconstruction years (approximately 1878-1900). Southern state legislatures in this period were characterized by two-party competition, and legislation remained relatively responsive to the preferences of both lower-income white and Black voters (Olson 2024). Furthermore, cross-racial class-based (“fusion”) coalitions had successfully formed to win control of state governments in Virginia (early 1880s) and North Carolina (mid-1890s), and had nearly won statewide elections in most other states (Alabama, Arkansas, Florida, Georgia, Louisiana, Tennessee, and Texas). These anti-elite populists and often cross-racial working-class movements were driven in part by demands to reduce regressive taxation and increase public spending (Kousser 1974; Hahn 2006; Hyman 1989).

Democratic Party elites sought a more “permanent solution” to these ongoing electoral threats (Perman 2003, p. 15). Satisfying populist demands by increasing progressive taxation and spending on whites was an unpalatable option (Hyman 1989). Furthermore, the continued reliance on informal repressive tactics was not entirely effective, as evidenced by the persistent voting of many Black citizens (Kousser 1999). Southern elites feared a “powerful fusion coalition of Populists and Republicans who were either on the verge of taking control of the state government or in power already” (Perman 2003, p. 323). They viewed the preservation of white supremacy through the disenfranchisement of Black voters as crucial for maintaining Democratic Party dominance and, by extension, their own

⁴Although assessments were unevenly applied—as was common nationwide—state revenues came primarily from progressive property taxes that fell disproportionately on landed elites in higher-Black-share counties (Jensen et al. 2024). Meanwhile, state funding for common schools rose well above prewar levels, and states introduced pensions for Confederate veterans.

hegemony. In a speech to the 1898 Louisiana Constitutional Convention, which adopted poll taxes and a literacy test, the delegate (and former Confederate senator) Thomas J. Semmes declared: “We met here to establish the supremacy of the white race, and the white race constitutes the Democratic party of the state. There is therefore, in my judgment, no separation whatever between the interests of the State and those of the Democratic party” (Perman 2003, 157).

Against this backdrop, Democrats across all fourteen states in our sample pursued measures to weaken Black voting power, succeeding in eleven. Between 1890 and 1906, these eleven states adopted a range of disenfranchising provisions through statutes, constitutional amendments, and new constitutions, often supplementing formal restrictions with informal voter suppression to secure the necessary majorities (Kousser 1974). Most states enacted several measures, including poll taxes, literacy tests, multibox laws, and secret ballot laws, that enabled local Democratic election officials to systematically disenfranchise African Americans (e.g., Greenberger 2022; Keele et al. 2021; Kousser 1974; Perman 2003; Valelly 2004).⁵ While these restrictions were not explicitly racial in nature, according to Gray and Jenkins (2024), their design and implementation, “reduced Black voting in the ex-Confederacy to near zero.” The voting rights of most lower-income and less educated Whites, on the other hand, were initially preserved through the adoption of “grandfather clauses” and other exemptions (Valelly 2004).⁶ From the adoption of these suffrage restrictions, the South largely remained a one-party system until the various Civil Rights laws of the 1960s.⁷

⁵Appendix Table A.1 reports these states, organized by the type of disenfranchisement measure and the date of adoption.

⁶Although these clauses were ultimately struck down by the Supreme Court (*Guinn v. U.S.*, 238 U.S. 347 (1915)), this decision was rendered after the conclusion of our study period.

⁷The Whites-only Democratic primary, adopted after Black disenfranchisement, gave low-income Whites greater voice and further eroded Black political power. As Caughey (2018) shows, however, it did not make state legislatures more responsive to mass White preferences—arguably because the apportionment of legislative seats, the channel we emphasize, was unaffected by the White primary.

3 Theoretical Framework

Our case lies at the intersection of three literatures in comparative political economy: 1) the relationship between franchise change and redistribution, 2) the effects of diversity and group conflict on public spending, and 3) the distributive consequences of malapportionment. While none explicitly examine how systematic outgroup disenfranchisement under conditions of malapportionment reconfigures redistribution within the enfranchised group, each offers useful benchmarks for thinking about the key variables at play in our setting.

The canonical franchise-redistribution logic holds that when poorer citizens gain the vote, the decisive voter's income falls below the mean, increasing demand for redistribution and expanding the tax-financed budget (Meltzer and Richard 1981). Evidence from the United States broadly aligns with this prediction: the removal of mid-twentieth-century voting barriers raised turnout among lower-income citizens and increased state transfers and welfare spending (Cascio and Washington 2014; Husted and Kenny 1997). Yet more recent studies show that the relationship between enfranchisement and spending can be nonmonotonic and context dependent (e.g., Aidt et al. 2022; Chapman 2024). Additionally, a complementary strand of research shows that the size of the electorate can shape the distributive calculus of politicians by shifting the composition of spending, even if its level remains unchanged: as the electorate broadens, narrowly targeted “club goods” become less electorally efficient and parties tilt toward broad public goods (Lizzeri and Persico 2004). Yet these accounts tend to neglect the role of diversity and rarely consider the effects of representational bias, typically treating vote shares rather than seats as the key drivers of party strategy. As a result, they provide little leverage for understanding how franchise reversals in racially diverse polities—under malapportionment—reshape patterns of intragroup redistribution.

A second relevant literature links ethnic and racial heterogeneity to redistribution and public goods, typically finding that diversity depresses universalistic spending and can enhance group-based exclusion (e.g., Alesina and Glaeser 2004; Habyarimana et al. 2007; Miguel and Gugerty 2005; Dahlberg et al. 2012). Within the U.S. South, historians and economists document how disenfranchisement diverted resources away from Black communities and widened racial inequalities in education (Margo

2007; Naidu 2012). Other work shows that the identity politics of (en)franchisement can reshape elite coalitions and state capacity: in South Africa, a cross-class White coalition sustained the implementation of high progressive taxes under Apartheid that increased redistribution (Lieberman 2003); in the Reconstruction and post Reconstruction South, enfranchising Black voters led to cross-class coalitions among Whites and elicited elite strategies to weaken state capacity and curtail redistribution (Suryanarayan and White 2021). These accounts illuminate intergroup redistribution and elite responses to racial threats, but they do not specify how institutional translation—in particular, malapportionment—mediates the fiscal consequences of disenfranchisement for intragroup (within White) redistribution.

Finally, a substantial literature on territorial representation examines how political geography and institutions jointly shape the allocation of public resources (e.g., Ansolabehere et al. 2002; Ansolabehere and Snyder 2008; Ardanaz and Scartascini 2013; Rickard 2018; Beramendi and Jensen 2019; Beramendi and Rogers 2022). This work demonstrates that malapportionment can significantly influence how centralized public resources are distributed across territorial units (e.g., from international, national, or sub-national governments to constituent parts). Yet it rarely considers how these representational biases interact with group-based politics or with expansions and contractions of the franchise.⁸

While existing accounts offer partial guidance for what to expect, they remain underdetermined because key variables relevant to our case fall outside their scope. Median-voter models suggest that removing poor Black voters shifts the electorate toward richer Whites, thereby reducing demand for redistribution and pushing overall social spending downward. Yet, without incorporating how legislative institutions translate votes into seats, these models provide little insight into the geographic

⁸Related research shows that legislators target large district groups (Jurado and León 2019), that geographically concentrated low-income groups receive more redistribution in single-member districts (Jusko 2015), and that politicians favor coethnics, though the extent and geography of this bias vary across contexts and goods (Burgess et al. 2015; Kramon and Posner 2013; Ejdemyr et al. 2018). Yet these perspectives omit crucial features of our case.

targeting of spending. Ethnic-favoritism theories, meanwhile, anticipate a reallocation of resources toward White-majority jurisdictions, but yield ambiguous expectations for overall redistribution—since in-group transfers may expand overall spending or merely replace outlays previously directed to the excluded. Additionally, none of these approaches specifies how outgroup disenfranchisement reshapes intragroup redistribution under conditions of malapportionment.

We address this gap by adopting what we consider the most parsimonious framework for capturing the interaction among key variables in our case—a standard minimum-winning-coalition (MWC) model, in which office-seeking actors assemble just enough support to secure a legislative majority and allocate side payments to pivotal voters to sustain it. Specifically, because legislative seats in the South were apportioned by total population rather than by the number of eligible or actual voters and because Black voters were spatially concentrated, disenfranchisement reduced and geographically concentrated the share of the White electorate needed to secure a legislative majority. This shrinkage in the MWC in terms of numbers and geography altered elites’ incentives; it allowed them to reduce broad, statewide redistributive efforts, while simultaneously shifting state-wide expenditures towards more elite-aligned goods and directing targeted transfers to a pivotal subset of Whites located in high Black-share areas. More importantly, and consistent with a MWC story, the fiscal effects were larger in states with higher Black population shares, which disproportionately decreased the numerical and geographical size of the MWC.

4 The Fiscal Consequences of Black Disenfranchisement

Trends in State Expenditures at the State Level

We begin by addressing our primary question on how Black disenfranchisement affected state government expenditures, especially on Southern Whites. While Black disenfranchisement likely had distinct and profound effects on local public spending (Kousser 1980), we focus exclusively on state government expenditures. We ultimately do so because, as we discuss in the next section, our proposed mechanism emphasizes how disenfranchisement interacted with existing state-legislative apportionment rules. Using an original annual dataset covering fourteen Southern states from 1880

to 1910, we first document broad spending patterns across major budget categories. This allows us to assess whether, after consolidating legislative power, elites redirected state resources away from goods broadly consumed by non-elite Whites and toward those that disproportionately benefited more affluent Whites. Next, in our main analysis, we examine how disenfranchisement influenced the distribution of state spending within states.

The dataset underlying our analysis was manually compiled from annual and biennial reports of state comptrollers, auditors, and treasurers. This extensive effort produced data for 403 out of a potential 434 state-year observations, covering total state expenditures and disaggregated spending across seven categories. Notably, no state is missing more than five years of data across the 31-year period. Our sample includes periods both before and after Black disenfranchisement in each state. To our knowledge, this dataset is the most comprehensive and fine-grained available for this era.

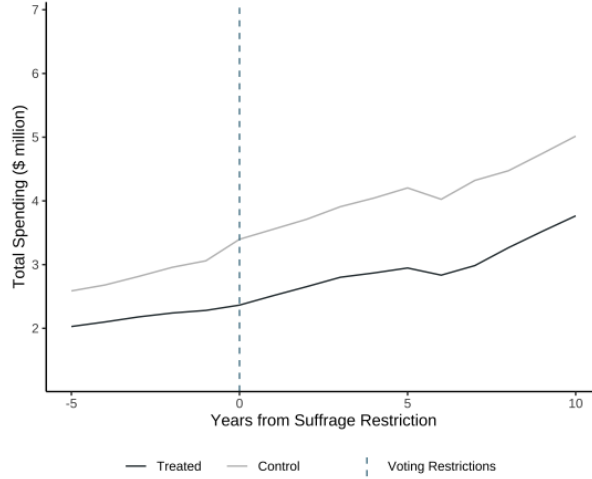
We begin by investigating median-voter expectations with total state outlays. To describe trends, Figure 1 aligns states by the year they adopted suffrage restrictions designating that implementation year as zero.⁹ The black line (treated states) and gray line (comparison group) move in parallel before and after adoption, suggesting little effect on total spending.

We next examine expenditure shares by category, classifying each good according to the incidence of its benefits and the incidence of its financing. To capture redistribution toward lower-income Whites, we focus on state spending on common schools and pensions for Confederate veterans. Common schools—the era’s principal redistributive good (Lindert 2004; Kousser 1980)—accounted for roughly 25% of state expenditures and were financed largely through progressive property taxes.¹⁰ Confederate pensions, which expanded substantially during this period, were likewise

⁹To construct these figures, we follow the method outlined in Paglayan (2022).

¹⁰Totals include state funding for Black common schools, although these expenditures disproportionately benefited White students, particularly toward the end of the studied period (Margo 2007). Poll taxes were often earmarked for schools but yielded only about one-fifth of the revenue raised by state property taxes dedicated to education (Jensen et al. 2024). We return to these issues later in the paper.

Figure 1: **Total State Spending (\$)** in States with and without Voting Restrictions



(a) Total State Spending

Notes: Average expenditures for treated states are aligned by restriction year; the comparison trend pools never- and not-yet-treated states in each event time.

funded through property taxes and targeted poor White veterans and widows.¹¹ Unlike state expen-

¹¹While Union veterans—including Southern Unionists and the Black soldiers of the U.S. Colored Troops—drew federal pensions, former Confederates were excluded from the federal system and hence depended entirely on their states. The programs began narrowly, as compensation for soldiers maimed or permanently disabled in the war; over the 1890s and 1900s most states dropped the disability requirement and extended coverage to indigent veterans and, increasingly, their widows, subject to means and residency tests. The pensions thus came to function less as disability compensation than as means-tested old-age and widows' relief for poorer Whites, and, consistent with elites' need for broad electoral support, they rose most where electoral competition from populists was most intense (Eli and Salisbury 2016). Because the programs were wholly state-financed, their generosity and eligibility varied substantially across states and over time: pension spending climbed from about 1% of state outlays in 1889 (the first suffrage restriction) to roughly 10% by 1910, exceeding 20% in four states. This secular expansion was common to counties within a state, however, and is therefore absorbed by our year fixed effects and state-specific trends; what our estimates recover is the within-state re-targeting of pensions across counties.

ditures on common schools (which were financed jointly by state and local governments and were enjoyed by non-poor Whites as well), Confederate pensions were a purely redistributive good funded solely by state governments. Appendix Figure A.1 shows a clear post-adoption gap for pensions and a modest divergence in school shares, with disenfranchising states spending less on average than untreated or not-yet-treated states.

We also examine five categories that did not disproportionately benefit lower-income Whites—higher education, prisons, debt service (including interest), courts, and insane asylums. Together, these categories accounted for more than 43% of state expenditures and were reported consistently across states from 1880 to 1910. Two clearly benefited narrow elite constituencies. Higher-education spending primarily supported universities and colleges serving a small, propertied White minority (Geiger 2014).¹² Prison expenditures—covering penitentiaries, convict-labor costs, and capture fees—served conventional penal functions but also supplied coerced labor that disproportionately benefited landed interests (e.g., Mancini 1996; Schwarz 2023; Lichtenstein 1996; Blackmon 2008). The distributional incidence of the remaining categories is less clear-cut. Debt service likely also favored elites, whereas courts and asylums were comparatively neutral.¹³ Appendix Figure A.2 shows that while higher education (a), prisons (b), and debt payments (c) rise faster in treated states, courts (d) and asylums (e) remained largely similar across states. Overall, disenfranchisement appears to re-allocate rather than reduce spending: broadly redistributive shares fall; narrowly targeted categories increase; others remain unchanged.

To assess whether these cross-state differences in spending trends reach statistical significance, we

¹²In 1910, fewer than 3% of American adults aged 24 and older held a college degree (*Statistical Abstract of the United States*, U.S. Census Bureau, 1999, Tables 265 and 1426).

¹³For our purposes, the key point is that debt service, courts, and asylums did not disproportionately benefit the poor, regardless of whether they were neutral or elite-aligned. Available evidence suggests that debt service favored elites, who prioritized payments to bondholders to preserve access to credit markets (Queralt 2022). The Readjuster Movement of the 1880s in Virginia supports this interpretation (Kousser 1974, 171–172).

report estimates using the Callaway and Sant’Anna (2021) staggered DiD estimator, which accommodates heterogeneous treatment effects across units. Results in Table 1 mirror the descriptive patterns. Panel A reports group-time effects weighted by group size (avoiding negative weights in TWFE) and shows little precise change in total spending (Model 1) but clear compositional shifts—declines in common school spending and a large, significant drop in pensions (about 0.75 SD), alongside increases in colleges, prisons ($\approx 50\%$), and debt service ($\approx 54\%$), with negligible effects on the judiciary and asylums; Panel B aggregates effects by treatment cohort to correct overweighting of early adopters and, while magnitudes differ slightly, corroborates the same pattern—no decline in overall budgets with a reallocation away from redistributive items (schools, pensions) toward narrowly targeted (and more elite-aligned) categories (colleges, prisons, debt service), including a notable $\sim 45\%$ rise in college spending. Figure A.3 confirms these patterns using spending shares as the outcome. Given the limited number of observations, we interpret the results with caution. Nonetheless, the evidence—including the event-study estimates in Figure A.4 and robustness checks showing that the timing of suffrage restriction adoption is uncorrelated with pre-treatment spending levels (Table A.2)—points to a significant reorientation of state spending following disenfranchisement.

Trends in State Expenditures at the County level

The documented state-level spending patterns show that disenfranchisement resulted in less overall redistributive spending by state governments on Southern Whites. We next examine whether this pattern holds for most Southern Whites within each state. We do so using county-level allocations of state spending for common schools and Confederate pensions from 1880-1910, compiled from auditors’, treasurers’, and superintendents’ reports. While we rely on never-treated states as a clean counterfactual for treated units in the state-level analysis, the sample in our county-level analysis is restricted to states that implemented disenfranchisement at some point.

We measure disenfranchisement with a binary indicator for whether a state had formal voting restrictions in place in a given year. This necessarily simplifies a complex set of institutions that

Table 1: **Effect of Suffrage Restrictions on State Expenditures: Average ATT Estimates**

Panel A <i>Simple Aggregation</i>	Total Spending	Schools	Pensions	Colleges	Prisons	Debt	Courts	Asylums
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post-Suffrage Restrictions	−0.194 (0.157)	−0.131 (0.129)	−0.174*** (0.050)	0.034 (0.021)	0.063** (0.031)	0.222** (0.106)	−0.028 (0.023)	−0.009 (0.025)
Observations	451	448	363	447	449	448	451	451
Panel B <i>Group Aggregation</i>	Total Spending	Schools	Pensions	Colleges	Prisons	Debt	Courts	Asylums
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post-Suffrage Restrictions	−0.117 (0.132)	−0.079 (0.115)	−0.136*** (0.030)	0.045** (0.018)	0.060** (0.025)	0.191** (0.081)	−0.019 (0.021)	0.004 (0.028)
Observations	451	448	363	447	449	448	451	451

Notes: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Models estimated using the Callaway and Sant’Anna (2021) estimator. The unit of analysis is the state-year. States are coded as treated after the first suffrage restriction is implemented. Standard errors are clustered at the state level. The control group includes not-yet-treated states.

varied across states in form, intensity, and enforcement.¹⁴ Our estimates should therefore be interpreted as the average effect of the institutional packages states adopted, not the marginal effect of any single restriction. This limits the scope of the estimand but does not itself threaten identification. Identification requires not that states adopted identical restrictions, but that treated and comparison units would have followed parallel trends absent formal disenfranchisement. We evaluate potential limitations of this approach in the robustness section (Tables A.9–A.11).

To capture how these rules altered counties’ political weight based on their racial composition, we interact this indicator with each county’s 1880 White population share.¹⁵ The interaction term between *Suffrage Restriction* and *Percent White* captures this under-representation dynamic: it equals

¹⁴Poll taxes differed in amount and whether they were cumulative; literacy tests varied in design, administrative discretion, and enforcement; and states often layered additional devices onto existing restrictions.

¹⁵We use the 1880 White-population share—the last major census before the onset of formal suffrage restriction—throughout our analysis to avoid endogeneity from post-treatment population changes.

zero for all counties before restrictions are adopted and, in the years following adoption, takes on the value of the county's 1880 White population share.

Our approach is rooted in a DiD design with multiple time periods and units, and a continuous treatment variable—White population share interacted with an indicator for suffrage restrictions:

$$Y_{cst} = \beta \text{Suffrage Restriction}_{st} \times \text{Percent White}_{cs\ 1880} + \lambda_t \times X_{cs\ 1880} + \lambda_t + \delta_c + \epsilon_{cst} \quad (1)$$

where Y_{cst} is per-capita state education or pension spending in county c in state s and year t . *Suffrage Restriction* _{st} indicates adoption;¹⁶ in state s at time t , while *Percent White* _{$cs\ 1880$} is the pre-treatment county White share. $X_{cs\ 1880}$ includes 1880 covariates—total population, urban population share, the total value of agricultural and manufacturing output, and assessed property value per White capita—each interacted with year fixed effects λ_t .¹⁷ County fixed effects δ_c account for unobserved, time-invariant county attributes, such as geographic, cultural, or political factors, and it absorbs the baseline level of *Percent White*; year fixed effects λ_t capture common shocks. Standard errors are clustered at the county level to account for potential autocorrelation over time. Our DiD design tests for an immediate directional shift in spending following Black disfranchisement, specifically whether counties with larger baseline White shares received relatively less state spending on redistribution. This is a demanding empirical test because fiscal policy is typically highly persistent, with expenditure patterns evolving only slowly over time.

Results

Table 2 estimates the effects of suffrage restrictions on per capita state allocations to counties for common schools. Column (1) includes only the restriction indicator, while column (2) adds pre-treatment controls—total population, urban population share, local economic output, and assessed property

¹⁶Because the severity of the poll tax and/or literacy test varied across counties and polling places due to variation in enforcement intensity, which our binary measure cannot capture, the resulting measurement error would likely attenuate our estimates.

¹⁷Each county-level measure was taken from the 1880 Census (Manson et al. 2024).

value per White capita in 1880—each interacted with year indicators. Results show that suffrage restrictions reduced per capita school funding. Columns (3) and (4) introduce an interaction between the restriction indicator and the 1880 White population share to capture heterogeneous effects. In counties with relatively low shares of White residents, disenfranchisement increased allocations to schools; in predominantly White counties, it led to retrenchment. Specifically, for counties with a near-zero White population share in 1880, suffrage restrictions increased per capita school funding by about seven and a half percent of a standard deviation, whereas counties that were entirely White saw a decrease of roughly one tenth of a standard deviation. As shown in Figure 2, the marginal effect turns from positive to negative once the White share exceeds 43 percent.

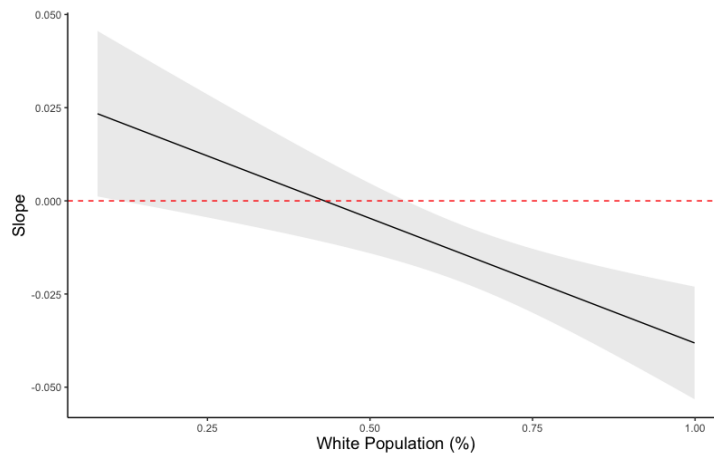
Table 2: Heterogeneous Effects of Disenfranchisement on per Capita State Allocations to Counties for Common Schools

	State Apportionment for Education per Capita			
	(1)	(2)	(3)	(4)
Suffrage Restriction	−0.022*** (0.004)	−0.014*** (0.004)	0.027** (0.013)	0.029** (0.013)
Suffrage Restriction × White Population 1880 (%)			−0.074*** (0.019)	−0.067*** (0.019)
Observations	25,638	22,286	24,530	22,286
R ²	0.857	0.959	0.937	0.959
County FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
State-specific time trends	✓	✓	✓	✓
Time-invariant covariates		✓		✓

Notes: The dependent variable measures per capita state educational funding across counties. All models account for year and county fixed effects, along with state-specific linear time trends. To ensure comparability, we restrict our sample to counties within states that implemented disenfranchisement at some point in time. The base term on White population share in 1880 is absorbed by the county fixed effects. The specifications in columns (2) and (4) include the following pre-treatment controls interacted with year indicators: county output, population size, urban-population share, and the assessed value of property per White capita in 1880. Additionally, columns (3) and (4) feature an interaction term between suffrage restrictions and the local White population share in 1880.

Although these results show heterogeneous effects, our per capita school measure certainly understates resources directed to Whites in high-Black-share areas. The dependent variable—total state

Figure 2: **Marginal Effects: The Moderating Role of White Pop. Share on the Relationship Between Disenfranchisement and per Capita State Allocations to Counties for Common Schools**



Notes: Marginal effects of disenfranchisement on per capita state educational funding across counties by White population share. The figure is based on results from model (4) in Table 2, which accounts for year and county fixed effects, along with state-specific linear time trends, and the following pre-treatment controls interacted with year indicators: county output, population size, the share of urban population, and the assessed value of property per White capita in 1880.

common-school revenues apportioned to each county divided by total county population—omits the large and growing racial gap in per-pupil spending. Margo (2007) reports that between 1890 and 1910 the statewide gap in teacher-salary spending per pupil between White and Black schools rose from 82% to 372%. Using both local and state accounts, Kousser (1980) likewise finds disproportionate gains for White schools in heavily Black North Carolina counties after disenfranchisement. Our estimates, which assume equal per-pupil spending across races, should therefore be viewed as a lower bound on the fiscal benefits to White students in high-Black-share counties.¹⁸

Next, we examine whether a similar county-level relationship holds for Confederate pension spending. Because pensions were reserved for White veterans and widows—unlike common school funds serving both Black and White students—we normalize expenditures by each county’s White population. Pensions also offer an analytical advantage: unlike school funding, they came solely from state governments, allowing us to rule out substitution between state and local education spending

¹⁸For more discussion of how our findings complement prior scholarship, see Appendix Section A.2.

(a concern addressed further below). The main limitation is data availability: complete county-level pension records exist for only seven states.¹⁹ Despite this constraint, the results in Table 3 and Figure A.5 reveal a similar pattern as with state common-school spending: disenfranchisement sharply reduced pensions in heavily White counties, while spending in high-Black-share counties held steady.

Table 3: Heterogeneous Effects of Disenfranchisement on Confederate Veteran Pension Spending per White Capita across Counties

	State Pension Spending per White Capita			
	(1)	(2)	(3)	(4)
Suffrage Restriction	−0.097*** (0.017)	−0.107*** (0.021)	−0.003 (0.023)	−0.008 (0.036)
Suffrage Restriction × White Population 1880 (%)			−0.165*** (0.039)	−0.174*** (0.048)
Observations	10,847	9,946	10,274	9,946
R ²	0.767	0.795	0.769	0.796
County FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Time-invariant covariates		✓		✓

Notes: The dependent variable measures state spending on Confederate pensions per White capita across counties. All models account for year and county fixed effects, along with state-specific trends. The sample includes only states that implemented disenfranchisement at some point in time. The base term on White population share in 1880 is absorbed by the county fixed effects. The specifications in columns (2) and (4) include the following pre-treatment controls interacted with year indicators: county output, population size, the share of urban population, and the assessed value of property per White capita in 1880. Additionally, columns (3) and (4) features an interaction term between suffrage restrictions and the local White population share in 1880.

Overall, the results reveal a sharp geographic divergence in the two main redistributive categories of the period. After disenfranchisement, residents of predominantly White counties received far less in education and pension funds than their counterparts living in counties with larger Black populations. Some education spending likely supported schools attended by elites themselves, meaning not

¹⁹While state totals are widely available, the amount allocated to each county was inconsistently reported over time or entirely omitted in four states. The seven states with complete county-level pension records over this period are AL, AR, FL, MS, NC, SC, and VA.

all funds were purely redistributive. Yet this possibility is also consistent with the broader finding that disenfranchisement allowed elites to channel more state resources towards their own priorities while not threatening their hold on power.

If Black disenfranchisement precipitated these shifts in state spending, it stands to reason that the effects should be more pronounced in states with larger Black populations for at least two reasons. First, disenfranchisement removed a greater share of the electorate, which alone could benefit Whites statewide. Second, high Black population states contained more counties with large Black majorities, creating sharp county-level heterogeneity in the proportion of voters excluded. By contrast, states with smaller Black populations had fewer such counties, limiting variation and dampening geographic disparities in voting power. For example, in 1880, the 25th percentile of White population share was 72% in Texas but only 28% in Mississippi, leaving far fewer counties in Texas with Black majorities. Appendix Table A.3 confirms this: while disenfranchisement reduced spending in high-White-share counties across all states, both its magnitude and its sensitivity to local demographics were much stronger in states with the highest Black-population shares (AL, FL, GA, LA, MS, and SC).

Robustness Checks and Threats to Inference

To assess the robustness of our results, we conduct a variety of supplementary analyses that address potential threats to inference, including model misspecification, confounding variables, selection, and treatment-effect heterogeneity. Full results appear in Appendix A; we summarize key findings here.

First, to rule out differential pre-trends, we re-estimate our models with state-specific quadratic trends (Table A.4). We then adopt a simplified specification including only county fixed effects to address recent concerns about the interpretability of two-way fixed effects estimators (Kropko and Kubinec 2020) (Table A.5). Results remain unchanged. We also replicate the analysis using the Sun and Abraham (2021) estimator, which accommodates heterogeneous effects and additional covariates. This approach enables us to verify the plausibility of the parallel trends assumption and to trace the dynamic effects of disenfranchisement on spending. The temporal patterns, presented in Figure A.6, closely track our main findings and reinforce a causal interpretation.

Next, we address the concern that the effects we observe may reflect the removal of poor White voters from the electorate rather than Black exclusion. Grandfather clauses help distinguish between these mechanisms because they were legal exemptions explicitly designed to preserve White suffrage. If poor White disenfranchisement were driving our results, the estimates should weaken or disappear once we restrict the analysis to states with grandfather clauses and interact post-adoption status with the local White population share. Instead, as Table A.6 shows, the results persist.²⁰ We further examine this possibility by following Olson’s (2024) approach: interacting disenfranchisement with both third-party vote shares and literacy rates. If suppressing third-party supporters or ensuring an educated electorate (rather than Black voter suppression) drove lower spending, these additional interactions should attenuate the main effect. They do not, however; the results remain robust (Table A.7).²¹

We also examine whether the adoption or timing of restrictions was correlated with state characteristics. Panel A of Table A.8 shows that static features—population size, output, White share, urbanization, and bureaucratic capacity—do not predict either the presence or timing of restrictions. Panel B presents two-way fixed effects models, likewise indicating that time-varying factors such as lynching also fail to predict when restrictions were adopted.

Our design compares each state’s spending before and after the date we code as the onset of disenfranchisement, which raises a concern: if a state had already begun curbing Black voting before that date, it was in effect partly treated beforehand, and the change we measure at the formal onset would understate the true effect. Several states fit this description. Georgia had levied a cumulative poll tax since 1877 and could be considered *always treated*; Alabama and Virginia had adopted secret-ballot laws; South Carolina had enacted the eight-box law; and North Carolina Democrats had passed an

²⁰The results also hold when we expand the sample to include states that adopted either grandfather clauses or understanding clauses, although this is a weaker test because understanding clauses protected White suffrage less reliably.

²¹Because local Populist strength was concentrated in high-White-share counties, third-party vote share and White population share are positively correlated. Including both interactions therefore pits the Populist and racial-composition accounts directly against each other, and only the White-share interaction remains significant—evidence that Populist intensity is not driving our results.

administrative election reform in 1889. To ensure these earlier measures do not drive our findings, we re-estimate excluding each such state individually and jointly, and control for the timing of these measures via interactions with White share; our results are unaffected (Tables A.9–A.11). Because these earlier restrictions attenuate our results, they also clarify what our estimates represent. The change we identify at the formal onset date captures the *additional* effect of entering a statewide disenfranchisement regime, rather than the cumulative effect of the broader transition from meaningful Black political participation to near-total exclusion. If, as seems plausible, informal suppression had already depressed Black participation before our coded onset, our estimates would understate the effect. A second consideration points the same way: insofar as Democratic elites remained unsure their newfound dominance would endure, they had reason to disperse benefits beyond a minimum-winning coalition, which would likewise attenuate the effects we detect.

We also examine whether broader constitutional reforms enacted alongside suffrage restrictions explain our findings. A detailed review of each state’s constitutional revisions reveals that with one possible exception—Alabama—concurrent reforms could not plausibly account for our results. Re-estimating our models with Alabama removed yields substantively identical results (see Appendix C).

Lastly, we consider whether the impact of disenfranchisement on school spending varied depending on the structure of education financing—specifically, whether school funds came primarily from state or local governments. In states where local governments played a dominant role in school funding (AR, LA, FL, and TN), effects are weaker, consistent with a more limited scope for state-level redistribution. As shown in Table A.12, effects are strongest where the state government was the main source of school funding. This suggests our education estimates may understate the impact of disenfranchisement on redistribution, as this was not a major form of redistributive state spending in four of the eleven states. By contrast, for Confederate pensions—funded entirely by state governments—we find a strong negative relationship between county White share and pension spending across all states. That is, the mechanism works as expected for strictly state-financed redistributive goods.²²

²²A final question is whether the greater education spending on Whites in high-Black-share counties

5 Political Control Before and After Disenfranchisement

The observed shifts in state government spending following the systematic disenfranchisement of Southern Blacks invite the obvious question of what accounts for these patterns. In this section we show that they follow from the interaction of Black disenfranchisement with the rules apportioning state legislative seats. Because those rules continued to count the disenfranchised Black population—or froze in place schedules that did—counties with large Black populations retained their seats while losing most of their voters, sharply reducing the share of Southern Whites the Democratic Party needed to control each legislature. While earlier work demonstrates the importance of this bias in sustaining rural elite power in the later Jim Crow era (e.g., Mickey 2015), we provide the first detailed and systematic account of its immediate effects.

The disenfranchisement of Black voters significantly reduced electoral threats to the dominance of rural elite-led Democratic Party in two key ways. First, scholars of the South have long recognized that since most Southern Blacks supported Republican or “fusion” parties, their removal substantially reduced the total statewide support for Democratic rivals—above all in the heavily Black plantation counties where most rural elites lived and where, as Appendix Figure B.1 shows, the Black population (25–61% of state totals in 1880) was concentrated (Jones et al. 2012).

Second, Black disenfranchisement allowed elites to control state legislatures—the branch responsible for public spending—without assembling broad statewide coalitions of White voters. It did so by changing how votes translated into legislative seats. Because county representation was based largely on total population rather than eligible voters²³, excluding Black voters increased the effective weight could be attributable to scale effects. Appendix Section A.3 discusses this possibility and explains why the evidence suggests it is unlikely.

²³The use of total population as the basis, along with the extension of the suffrage to the formerly enslaved, were reforms implemented during Reconstruction. Minor exceptions to this basis include several state constitutions that mandated a minimum of one representative per county in their lower house (Alabama, Florida, North Carolina, and South Carolina), thereby distorting representation from a strictly “one person, one vote” principle. Additionally, South Carolina retained its highly dis-

of White votes in high-Black-share counties: the smaller the White share of the population, the greater the weight of each White vote. As we show, the resulting overrepresentation of these counties enabled a small minority of each state's White population to control the legislature.

To demonstrate this effect, we focus on the apportionment of representation in the two legislative chambers of each state. Specifically, we calculate the proportion of White residents living in counties with the highest Black populations whose apportionment of seats was sufficient to guarantee a majority of seats.²⁴ The results, presented in Table 4, are reported separately by state and legislative chamber: columns 1 and 2 for the lower house, and columns 3 and 4 for the upper house. Given substantial demographic heterogeneity among Southern states, we present our results separately for three distinct state groupings. First, as reported in panel A, we pool counties from the six "Deep South" states where Blacks represented at least 40% of the population (AL, FL, GA, LA, MS, and SC). Panel B includes counties from the three states with moderate Black populations (25-40%, AR, NC, and VA). Finally, Panel C contains counties from the two states below 25% (TN and TX).

The first column presents the proportion of each state's White population residing in the highest Black-share counties that, together, held a majority of seats in the lower house. For example, in Alabama's House of Representatives, Democrats could secure a majority of seats by winning counties that contained only 29% of the state's White population. These winning-coalition counties had Black-population shares ranging from 40% to 88%. This range is reported for each state's lower chamber in column 2. In each Deep South state, Democrats could secure legislative majorities in their lower chamber relying exclusively on counties where the Black population share exceeded 40%. In two additional states (NC and VA), Democrats could achieve a majority by carrying counties with at least 31% Black share. In the other states, Democrats would need majority support in counties with much lower Black population shares. Columns 3 and 4 report analogous results for each state's upper chamber, demonstrating that, in most states, securing an upper-house majority required winning majorities in

tortionary prewar system of allocating one senator per county.

²⁴This calculation is based on the apportionment reflected in either state constitutions or statutes stipulating county representation as of 1910.

Table 4: **State Legislature Majority in High Black-Population Share Counties, 1910**

	Lower House		Upper House	
	Share of State's White Pop. in Winning Coalition Counties (%)	Range of Black-pop. Share in Winning Coalition Counties (%)	Share of State's White Pop. in Winning Coalition Counties (%)	Range of Black-pop. Share in Winning Coalition Counties (%)
A) High Black-pop. Share State (>40%)				
Alabama	29	(40, 88)	27	(42, 88)
Florida	39	(40, 76)	37	(41, 76)
Georgia	36	(48, 86)	33	(50, 86)
Louisiana	34	(43, 92)	36	(42, 92)
Mississippi	30	(55, 94)	28	(56, 94)
S. Carolina	37	(59, 87)	34	(61, 87)
Avg. High Black-pop. State				
B) Mid Black-pop. Share State (25-40%)				
Arkansas	39	(17, 85)	36	(19, 85)
N. Carolina	40	(31, 65)	38	(33, 65)
Virginia	40	(35, 74)	39	(36, 74)
Avg. Mid Black-pop. State				
C) Low Black-pop. Share State (<25%)				
Tennessee	38	(21, 75)	38	(21, 75)
Texas	38	(18, 64)	38	(18, 64)
Avg. Low Black-pop. State				
D) Non-Disenfranchising				
Kentucky	46	(10, 49)	45	(10, 49)
Maryland	31	(15, 52)	20	(24, 52)
Missouri	62	(3, 20)	55	(4, 20)
Avg. Non-Disenfranchising State				

Notes: Columns 1 and 3 show the proportion of each state's total White population residing in the highest Black-share counties that collectively held a majority of seats in the state's lower (Column 1) and upper (Column 3) houses in 1910. Columns 2 and 4 report the lowest and highest Black-population shares among these winning-coalition counties in the lower (2) and upper (4) chambers. The disenfranchising states include the 11 Southern states that implemented literacy tests and/or poll taxes. Sources: Kousser (1974); Valelly (2004); US Census, 1910. Legislative representation was manually coded from the relevant statutes or constitutional provisions specifying the apportionment for each chamber of each legislature circa 1910.

a set of counties containing an even smaller proportion of each state's White population.

Importantly, while these winning-coalition counties contained only a minority of each state's White population, even these figures, low as they may seem, substantially overstate the level of White support Democrats actually needed to secure legislative control (likely more than twice what was required in practice). Under the first-past-the-post electoral system, the Democratic Party needed

only a simple majority of voters within each district.

We further show in Appendix B that apportionment was not biased prior to Black disenfranchisement. While many scholars have recognized that malapportionment was a pivotal element of rural elite political strength during Jim Crow (Key 1984; Mickey 2015), we are the first to our knowledge to show that this emerged as a result of democratizing reforms enacted during Reconstruction combined with Black disenfranchisement. This analysis also rules out concerns that malapportionment existed prior to the formal disenfranchisement of Southern Blacks.

Conclusion

This paper examines how the Jim Crow-era disenfranchisement of Southern Blacks reshaped the fiscal state—and, crucially, *which* Whites gained and which lost. Using a new annual panel of disaggregated state spending for fourteen Southern states (1880-1910) and original measures of representational bias, we show that removing Black voters from the electorate while continuing to count them in the apportionment base contracted and geographically concentrated the minimum winning coalition of Whites that rural elites needed to command legislative power. Aggregate outlays did not fall; instead, budgets were reoriented away from redistributive items and toward narrowly targeted or elite-aligned spending categories—higher education, prisons, and, plausibly, debt service—while the remaining redistribution was directed to Whites residing in high-Black-share counties. In short, disenfranchisement redirected the state rather than shrinking it, with gains accruing to a geographically pivotal subset of Whites.

The mechanism is institutional. Legislative seats remained apportioned by *total* population under rules inherited from Reconstruction-era constitutions, so Black disenfranchisement overrepresented White voters in precisely those counties with the largest Black populations. Using our *Pivotal White Share* measure, we show that a legislative majority could thereafter be secured by winning pluralities in counties containing only 27–40% of the statewide White population, concentrated in the highest-Black-share areas. Shrinking and localizing the winning coalition weakened incentives for broad, statewide redistribution and encouraged targeted side-payments to pivotal Whites alongside greater

spending on elite-aligned categories.

Interpreting our results requires attention to both scope conditions and the limits of the data. First, we ask how disenfranchisement and apportionment rules reallocated political power *among Whites*; we therefore do not assess the *direct* welfare consequences for Southern Blacks, an important question explored by others (e.g., Althoff and Reichardt 2024; Naidu 2012; Margo 2007). Second, because formal restrictions consolidated an exclusion that informal suppression had already begun in several states, our estimates capture entry into a formal disenfranchisement regime and are best read as conservative. Third, because our mechanism runs through the *state* legislature, our outcomes are state rather than combined state and local expenditures; disenfranchisement likely shaped local spending as well, but those channels—particularly district-level school finance—lie beyond our scope.²⁵

The paper advances three debates. First, it offers a comprehensive portrait of the fiscal consequences of Black disenfranchisement for Southern Whites. Rather than simple retrenchment, we find a compositional shift:²⁶ less average redistribution to Whites statewide, coupled with geographically targeted redistribution to Whites in high-Black-share counties and higher spending on elite-oriented goods. These facts temper median-voter expectations and refine classic and contemporary claims linking (dis)enfranchisement to the size of government (Meltzer and Richard 1981; Acemoglu et al. 2015; Scheve and Stasavage 2017). Where representation is biased and the governing coalition small and spatially concentrated, suffrage restriction need not shrink the state; it redirects it toward elite uses and narrow side-payments. More generally, the interaction we study—out-group disenfranchisement under apportionment rules that count the disenfranchised—has broader comparative resonance: as Bateman (2018, p. 6) notes, the conjunction of democratization and disenfranchisement is neither rare nor uniquely American but “the result of deliberate choices made by elite political coalitions looking to gain and hold on to power.”

²⁵We do, however, address complexities around public education, which is jointly financed by state and local governments.

²⁶This echoes findings from U.S. suffrage *expansions*, where reforms often reallocated spending rather than expanding total outlays (Cascio and Washington 2014; Husted and Kenny 1997).

Second, we bridge and qualify literatures on diversity and public goods, franchise change, and malapportionment by showing that fiscal consequences hinge not only on *who* votes, but on how votes are translated into *seats*. Prior research demonstrates that (dis)enfranchisement shapes intra-White, cross-class coalitions that can facilitate or block redistribution (Lieberman 2003; Suryanarayan and White 2021); we extend this insight by tracing the institutional channel that makes those coalitions pivotal. In the Southern case, disenfranchisement reduced the share of lower-income Whites needed to sustain elite control, weakening incentives for broad redistribution while encouraging targeted transfers to pivotal Whites. More generally, wherever out-group disenfranchisement coexists with population-based apportionment, the demographic-to-seat translation reconfigures governing coalitions and, with them, the map of intragroup redistribution.

Third, we contribute to the study of the political economy of the “Solid South” by identifying a fiscal logic that helped sustain one-party dominance: elites preserved hegemony by coupling targeted transfers to strategically over-represented Whites with a broader reallocation of spending toward elite priorities. This logic resolves three puzzles: why aggregate spending did not contract despite the reduced franchise; why Southern schooling continued to lag national standards even as state funds were redirected toward Whites in high-Black-share counties—a reallocation that was relative rather than absolute, and confined to a small share of the White population; and why state legislatures answered to so restricted a constituency even as Southern members of Congress and statewide officeholders grew increasingly responsive to mass White opinion (Caughey 2018)—a disjuncture our *Pivotal White Share* traces to apportionment rules that still counted the disenfranchised. Considered alongside evidence that the incidence of Southern state taxes in this period fell primarily on the landed elite (Jensen et al. 2024), the budgetary reorientation we document aligns with fiscal-contract views of the state (e.g., Bates 1983; Levi 1988; North and Weingast 1989; Timmons 2005): elites maintained support through selective benefits while preserving the capacity to finance their preferred public goods. More broadly, theorizing democratic reversal under malapportionment should proceed from *seats*, not votes: when institutional translation shrinks and concentrates the coalition, fiscal policy becomes the currency of elite power persistence. That is the lesson of the Solid South we document here.

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Supplemental Information

The Fiscal Consequences of Black Disenfranchisement during Jim Crow for Southern Whites

Contents

A Supplementary Data and Analysis	page
Table A.1: Southern States: Year and Type of Black Disenfranchisement	1
Section A.1: Black Disenfranchisement and State Spending	1
Figures A.1: Suffrage Restrictions and Redistributive Spending	2
Figures A.2: Suffrage Restriction and Other Spending Categories	3
Section A.2: Differences from Previous Research	4
Figure A.3: Average ATT Estimates: Share of Spending Allocated to Various Categories	4
Figure A.4: Event-Study Estimates: Suffrage Restrictions and State Spending by Category	5
Table A.2: Pre-Treatment Outcome Levels, by Cohort (thousand \$)	6
Section A.3: Scale-Cost Alternative Explanation	7
Table A.3: Variation by State-Level Black Population Share	8
Table A.4: Accounting for Quadratic Time Trends	9
Table A.5: County Fixed Effects	10
Figure A.5: Marginal Effects Wh. Pop. Sh and Pension Spending	11
Figure A.6: Event-Study Analysis: Variation by Counties' White Population Share	12
Table A.6: States with Grandfather Clause	13
Table A.7: Accounting for Literacy and Third-Party Vote	14
Table A.8: Predictors of Disenfranchisement Timing	15
Table A.9: Accounting for the Secret Ballot	16
Table A.10: Accounting for 8-Box Law	17
Table A.11: Accounting for Secret Ballot and 8-Box Laws, and Excluding GA and NC	18
Table A.12: Variation by State Financing Structure	19
B Black Disenfranchisement and Statewide White Political Control	19
Figure B.1: Black Population Share across Counties, 1880	20
Table B.1: County Relative Representation (RRI), 1880 and 1910	23
C Alternative Explanations	23
Figure C.1: Effect of Suffrage Restrictions on College Expenditures, excluding Alabama	25
References	25

Appendix A: Supplementary Data and Analysis

Table A.1: **Southern States: Year and Type of Black Disenfranchisement**

	Black Population Share 1880 (%) (1)	Type of Suffrage Restriction (2)	Year Enacted (3)
Alabama	47.5	LT, PT	1900
Arkansas	26.3	PT	1891
Florida	47.0	PT	1890
Georgia	47.0	LT, PT	1906*
Kentucky	16.5	None	
Louisiana	51.5	LT, PT	1896
Maryland	22.5	None	
Mississippi	57.5	LT, PT	1890
Missouri	6.7	None	
N. Carolina	38.0	LT, PT	1900
S. Carolina	60.7	LT, PT	1893
Tennessee	26.1	PT	1890
Texas	24.7	PT	1900
Virginia	41.8	LT, PT	1902

Notes: LT = Literacy Test. PT = Poll Tax. *Georgia enacted the literacy test in 1906 and poll tax in 1877. Sources: Kousser (1974); Valelly (2004), Walton et al. (2012, Table 13.9); US Census (1880).

A.1 Black Disenfranchisement and State Spending

Event Study. To test for parallel trends and evaluate dynamic treatment effects in the post-treatment period, we present the Callaway and Sant’Anna (2021) event study estimates, which account for treatment effect heterogeneity. These figures support the plausibility of the parallel trends assumption, as they reveal no discernible pre-trends across any of the dependent variables.

Figure A.4 plots treatment effects from five years before to ten years after disenfranchisement. Echoing earlier results, voting restrictions had no clear impact on total spending (a) or common school funding (b), with all ATT estimates near zero. In contrast, pension spending (c) declined significantly, with nearly all group-specific average treatment effects negative and statistically significant.

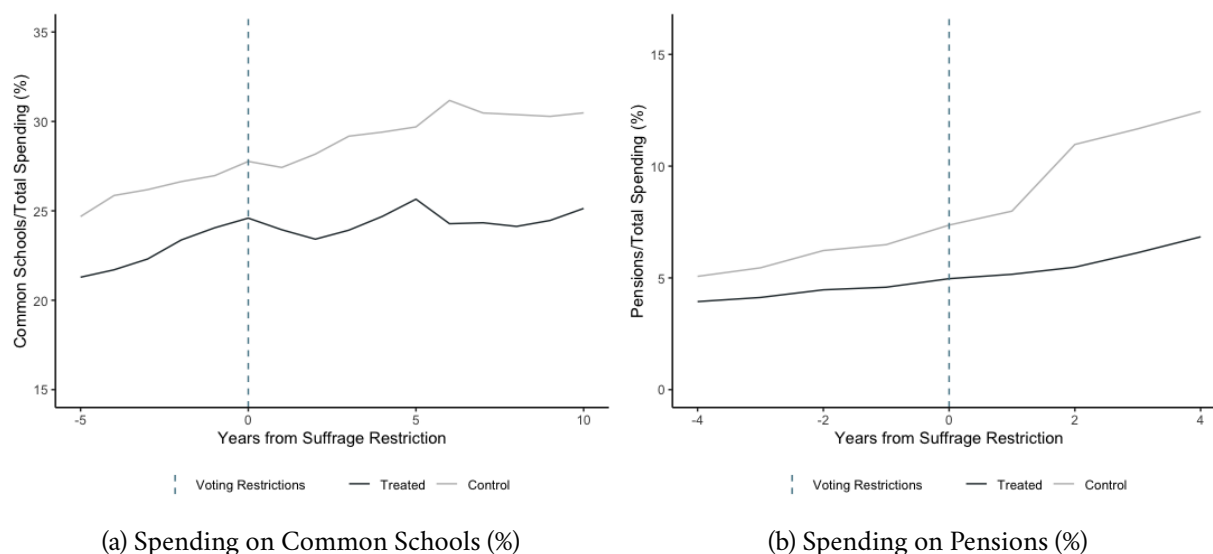
Non-redistributive spending shows varied effects. Expenditures on White colleges (d), prisons (e), and debt service (f) rose after disenfranchisement, with most estimates positive and several significant. Courts (g) and asylums (h) show no meaningful change, with estimates near zero. The results

using spending shares are broadly consistent, with the exception of debt payments, which show no significant change.¹

Assessing Pre-Treatment Differences in Fiscal Outcomes. We also address concerns that states' characteristics might systematically influence the timing of suffrage restriction measures. Beyond exploring potential differences in these characteristics, it is also important to determine whether states adopting suffrage restrictions at different times displayed noticeable pre-treatment disparities in fiscal outcomes. Although the main requirement for a difference-in-differences design is that counter-

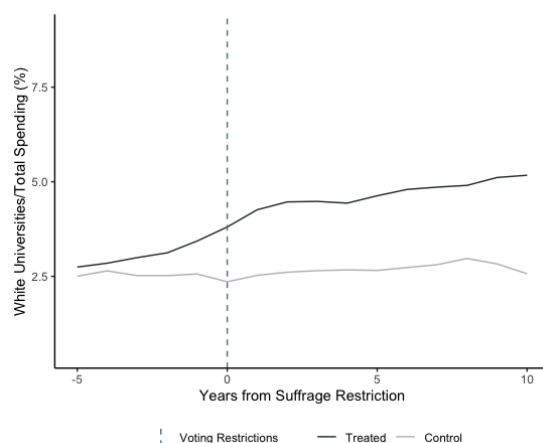
¹Due to space constraints, figures showing cohort-specific effects for both absolute amounts and spending shares are not included here but are available upon request.

Figure A.1: Redistributive Spending (% of State Total) in States with and without Voting Restrictions

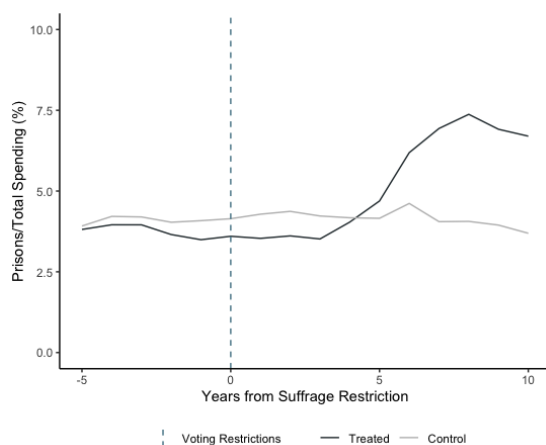


Notes: The black lines represent the average share of state expenditures on (a) common schools and (b) pensions in disenfranchising states. In Figure (a), the comparison group encompasses both states that never adopted these restrictions and those that had not yet done so. The counterfactual trend, represented by the gray line, is calculated by averaging the expenditures of all states without voting restrictions for each year a particular state implemented such measures. These averages are then aggregated according to the number of years before and after disenfranchisement. In Figure (b), the comparison group is limited to not-yet-treated states only, as the never-treated states did not record pension spending during the period under study. As a result, the post-treatment period in this figure is shorter, reflecting the number of years that elapsed between the implementation of suffrage restrictions in the last two states to adopt such measures.

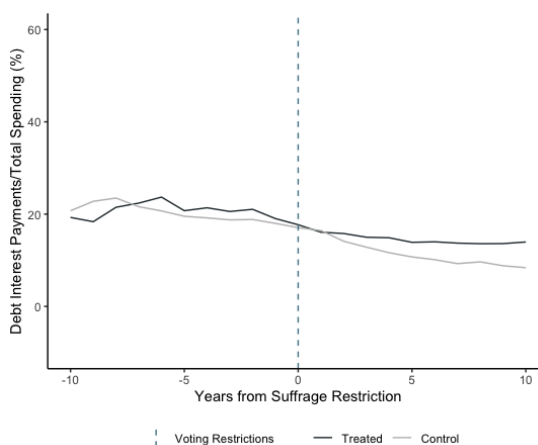
Figure A.2: Other Expenditure Categories (% of State Total) in States with and without Voting Restrictions



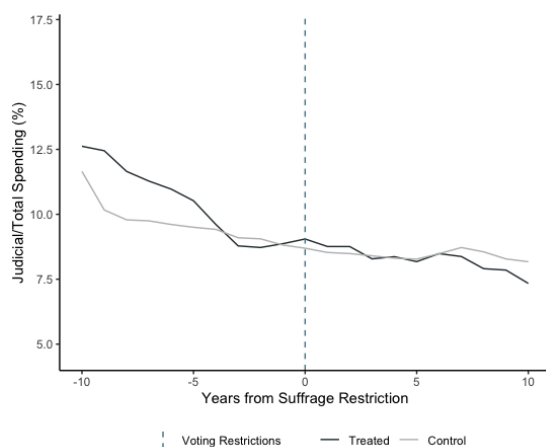
(a) Spending on White Higher Education (%)



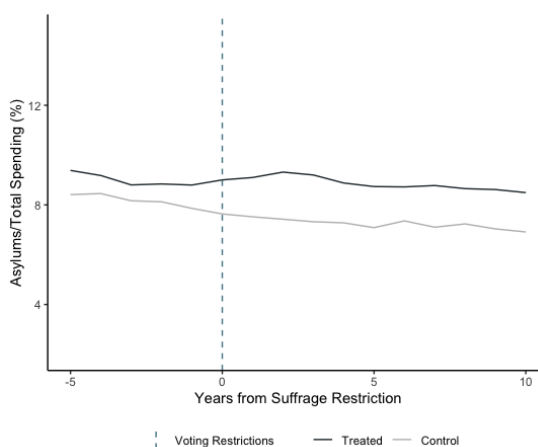
(b) Spending on Prisons (%)



(c) Spending on Debt & Interest Payments (%)



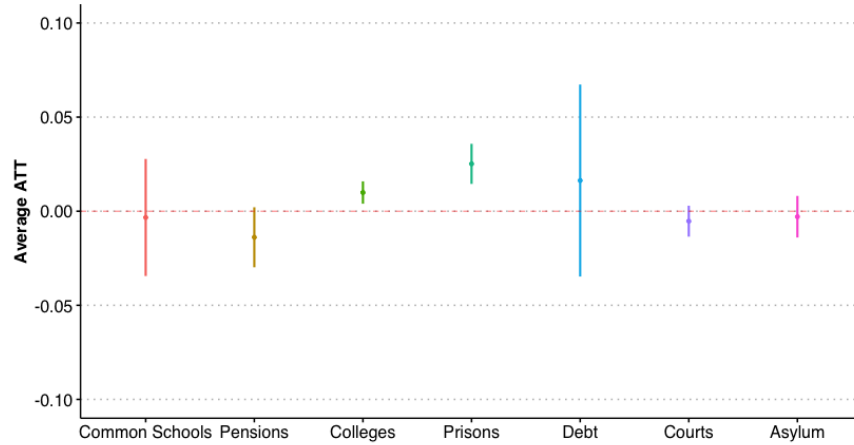
(d) Spending on Courts (%)



(e) Spending on Asylums (%)

Notes: The black lines show average share of spending on (a) white higher education, (b) prisons, (c) debt and interest payments, (d) courts, and (e) insane asylums across disenfranchising states. The gray lines reflect the average trend of the comparison group, which includes both never-treated and not-yet-treated states.

Figure A.3: **Average ATT Estimates: Share of Spending Allocated to Various Categories**



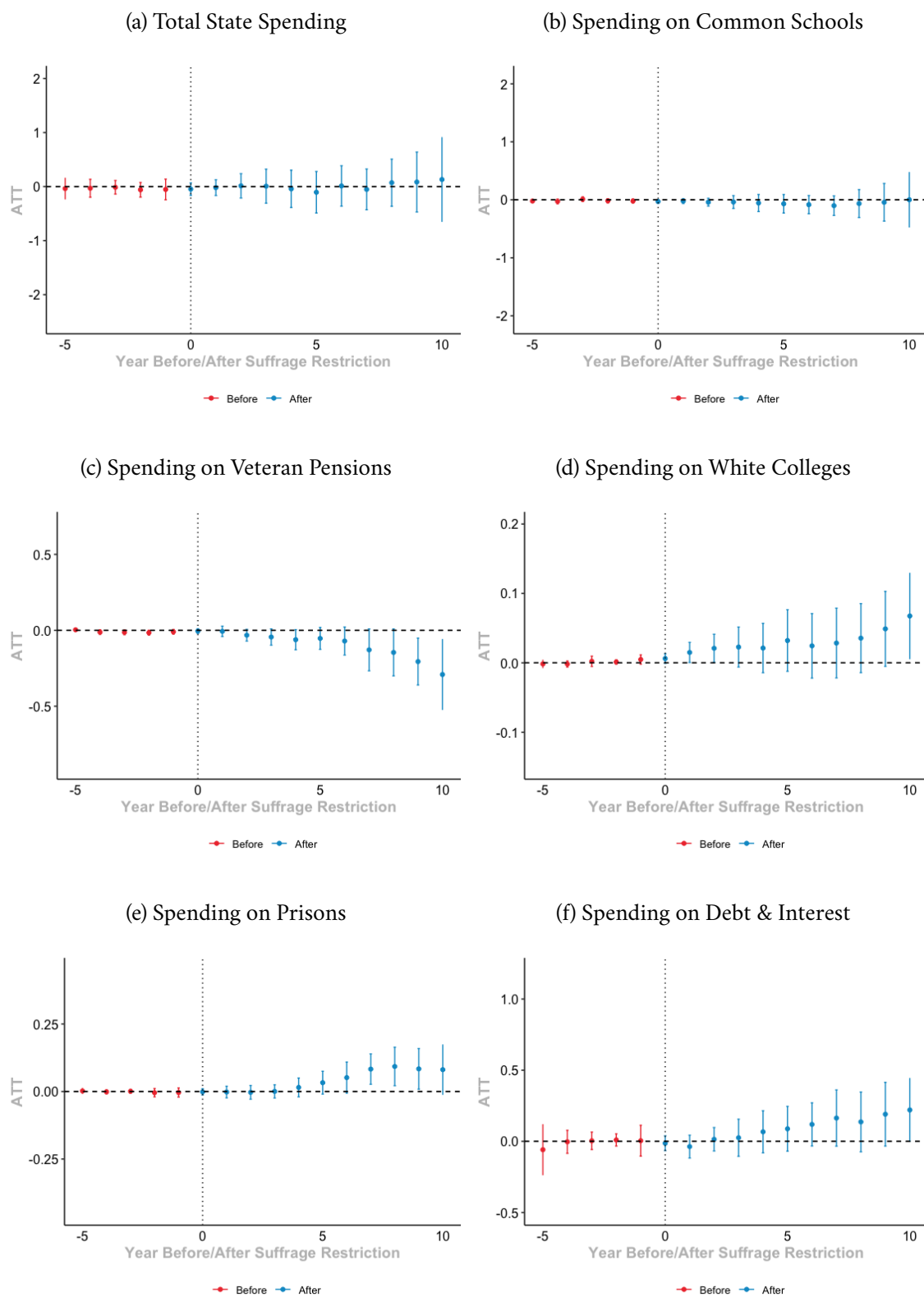
Notes: Estimates and 95% confidence intervals of the average treatment effect on the treated (ATT) for suffrage restrictions, aggregating all cohort-specific effects. Outcomes include the share of spending allocated to common schools, pensions, colleges, prisons, debt service, court systems, and asylums. All estimates are calculated using Callaway and Sant'Anna (2021) doubly robust estimation method.

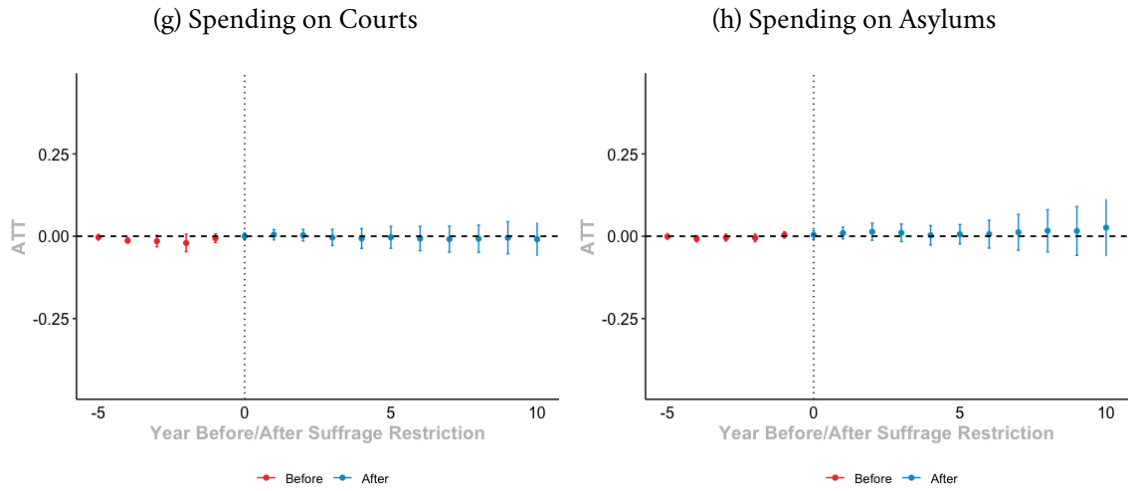
factual trends be parallel across groups, substantial pre-treatment differences in outcome levels can still pose identification concerns. To investigate this issue, Table A.2 reports average pre-treatment spending levels across treatment cohorts (columns 1-7) and tests for systematic differences using unconditional F-tests (column 8). The results show that the timing of suffrage restriction adoption is not correlated with pre-treatment spending levels, suggesting that significant level differences are unlikely to bias our estimates.

A.2 Differences from Previous Research

A number of economic historians (Kousser 1980; Margo 1982, 1984, 2007; Pritchett 1989) have examined how disenfranchisement shaped the Southern public school systems. Their work builds on the early arguments of Bond (1934) and Harlan (1969), who suggested that Whites in heavily Black counties diverted public school funds designated for Black students to avoid higher taxes yet still provide adequate schooling for white children. Although this work documents how White educational spending increases with the Black share of the population, the prevailing assumption is that this reflects primarily the diversion of funds away from Black schools (Gerber 1991), leaving the rest

Figure A.4: Event-Study Estimates: Effects of Suffrage Restrictions on Total State Spending and Categories Based on Length of Exposure to Treatment





Notes: Event time estimates and 95% confidence intervals for spending on (a) total state spending, (b) common schools, (c) pensions, (d) colleges, (e) prisons, (f) debt interest payments, (g) courts, and (h) insane asylums (million \$). Dynamic ATTs are calculated based on 5 pre- and 10 post-treatment years. The sample includes 14 Southern states, of which three are never-treated. The sample for Figure (c) encompasses 11 eventually-treated states. Standard errors clustered at the state level. All estimates calculated using Callaway and Sant’Anna (2021) doubly robust estimation method.

Table A.2: Pre-Treatment Outcome Levels, by Cohort (thousand \$)

	1890 Cohort	1891 Cohort	1893 Cohort	1896 Cohort	1900 Cohort	1902 Cohort	1907 Cohort	F-Test P-value
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Total Spending	1026.73	1374.29	1080.34	1504.91	1841.06	2931.91	1441.34	0.879
Spending on Schools	157.84	213.72	0.00	170.96	776.04	714.39	309.79	0.843
Spending on Pensions	0.96	0.00	0.00	0.00	24.57	0.00	0.00	0.943
Spending on Prisons	5.23	48.86	67.63	0.00	90.70	31.58	5.04	0.714
Spending on Colleges	30.63	14.77	17.50	13.75	54.24	56.50	8.00	0.864

Notes: The table presents mean outcome levels for states that implemented suffrage restrictions in 1890, 1891, 1893, 1896, 1900, 1902, and 1907. All variables are measured in 1885. Column 8 displays the p-value of unconditional F-Tests, which assess the equality of means across all seven groups of states.

of the White South unaffected by disenfranchisement. By contrast, our findings show that although Whites in high-Black-share areas did benefit from diverted school funds, they also gained from an increasingly disproportionate allocation of state revenues to these same regions, while funding for high-White-share counties fell. Our findings thus highlight that disparities in school spending rose not only between Black and White students in high-Black-share counties but also among Whites across different parts of each state. This pattern helps explain why Southern education standards overall remained well below national levels: a small, privileged minority could secure better schools while much of the region's White population lagged behind. Illustrating this, Gerber (1991) finds that in four planter states, the top decile of counties in Black population share spent more on White education than the non-Southern average—yet these counties made up only 3 percent of the White population.

A.3 Scale-Cost Alternative Explanation

An alternative interpretation of our findings suggests that segregated schools in areas with few White students were mechanically more expensive to operate due to scale inefficiencies. However, this explanation cannot account for the *change* in the relationship between racial composition and spending that we identify: our difference-in-differences design absorbs any time-invariant cost differences across counties through county fixed effects. The relevant question is why the gap in per capita spending between high- and low-White-share counties *widened* after disenfranchisement—a change in the differential that mechanical cost differences cannot explain. Moreover, the same geographic pattern appears in Confederate pension allocations (Table 3, Figure A.5), a purely state-funded transfer to individuals with no connection to the fixed costs of operating segregated schools. The pension results thus help rule out mechanical cost explanations and point instead to a political reallocation of state resources.

Table A.3: **Variation by State-Level Black Population Share: Heterogeneous Effects of Disenfranchisement**

Panel A: State Apportionment for Education per Capita		
	State Apportionment for Education	
	Low Black Share States (1)	High Black Share States (2)
Suffrage Restriction	0.021 (0.022)	0.076*** (0.014)
Suffrage Restriction $\times$ White Pop. 1880 (%)	-0.052* (0.030)	-0.088*** (0.023)
Observations	12,385	9,901
R ²	0.970	0.909
County FE	✓	✓
Year FE	✓	✓
State-specific trends	✓	✓
Time-invariant covariates	✓	✓

Panel B: Pension Spending per White Capita		
	Pension Spending per White Capita	
	Low Black Share States (1)	High Black Share States (2)
Suffrage Restriction	0.037 (0.026)	0.117** (0.048)
Suffrage Restriction $\times$ White Pop. 1880 (%)	-0.029 (0.039)	-0.352*** (0.082)
Observations	5,206	4,740
R ²	0.859	0.804
County FE	✓	✓
Year FE	✓	✓
State-specific trends	✓	✓
Time-invariant covariates	✓	✓

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel A reports effects on per capita state education funding; Panel B on pension spending per White capita. All models include year and county fixed effects, state-specific linear trends, and are limited to counties in states that enacted disenfranchisement. Column (1) includes states with lower Black population shares—AR, TN, TX, NC, VA (Panel A); AR, NC, VA (Panel B). Column (2) includes states with higher shares—AL, FL, GA, LA, MS, SC (Panel A); AL, FL, MS, SC (Panel B), based on 1910 population. All models control for 1880 county output, population, urban share, and property value per White capita, each interacted with year indicators.

Table A.4: **Accounting for Quadratic Time Trends: Heterogeneous Effects of Disenfranchisement**

Panel A: State Apportionment for Education per Capita		
	State Apportionment for Education per Capita	
	(1)	(2)
Suffrage Restriction	0.033*** (0.012)	0.036*** (0.012)
Suffrage Restriction $\times$ White Population 1880 (%)	−0.074*** (0.019)	−0.057*** (0.018)
Observations	24,530	22,286
R ²	0.944	0.967
County FE	✓	✓
Year FE	✓	✓
State-Specific Linear Time Trends	✓	✓
State-Specific Quadratic Time Trends	✓	✓
Time-invariant covariates		✓

Panel B: Pension Spending per White Capita		
	State Pension Spending per White Capita	
	(1)	(2)
Suffrage Restriction	0.071*** (0.024)	0.073** (0.029)
Suffrage Restriction $\times$ White Population 1880 (%)	−0.095** (0.037)	−0.108** (0.047)
Observations	10,274	9,946
R ²	0.817	0.839
County FE	✓	✓
Year FE	✓	✓
State-Specific Linear Time Trends	✓	✓
State-Specific Quadratic Time Trends	✓	✓
Time-invariant covariates		✓

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel A shows results for per capita state education funding by county; Panel B for pension spending per White capita. All models include county and year fixed effects, plus state-specific linear and quadratic trends, and are restricted to counties in states that enacted disenfranchisement. The 1880 White population share is absorbed by county fixed effects. Column (2) adds pre-treatment controls—county output, population, urban share, and property value per White capita—each interacted with year indicators.

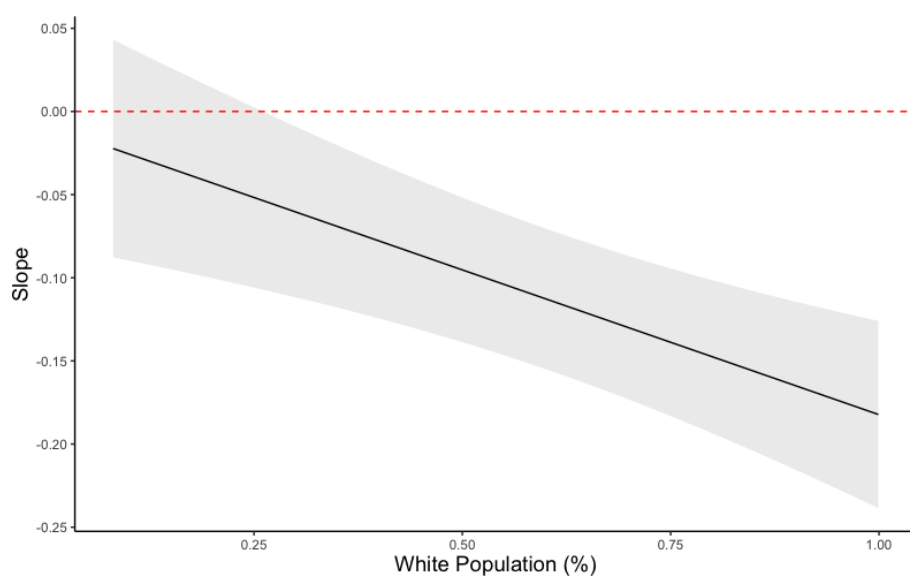
Table A.5: **County Fixed Effects: Heterogeneous Effects of Disenfranchisement**

Panel A: State Apportionment for Education per Capita				
	State Apportionment for Education per Capita			
	(1)	(2)	(3)	(4)
Suffrage Restriction	−0.028*** (0.005)	−0.013*** (0.004)	0.035*** (0.012)	0.044*** (0.012)
Suffrage Restriction × White Pop. 1880 (%)			−0.090*** (0.019)	−0.089*** (0.018)
Observations	25,638	22,286	24,530	22,286
R ²	0.845	0.948	0.925	0.948
County FE	✓	✓	✓	✓
State-specific trends	✓	✓	✓	✓
Time-invariant covariates		✓		✓

Panel B: Pension Spending per White Capita				
	Pension Spending per White Capita			
	(1)	(2)	(3)	(4)
Suffrage Restriction	−0.044*** (0.014)	−0.040** (0.016)	0.039* (0.023)	0.049 (0.034)
Suffrage Restriction × White Pop. 1880 (%)			−0.142*** (0.041)	−0.151*** (0.053)
Observations	10,847	9,946	10,274	9,946
R ²	0.732	0.748	0.732	0.749
County FE	✓	✓	✓	✓
State-specific trends	✓	✓	✓	✓
Time-invariant covariates		✓		✓

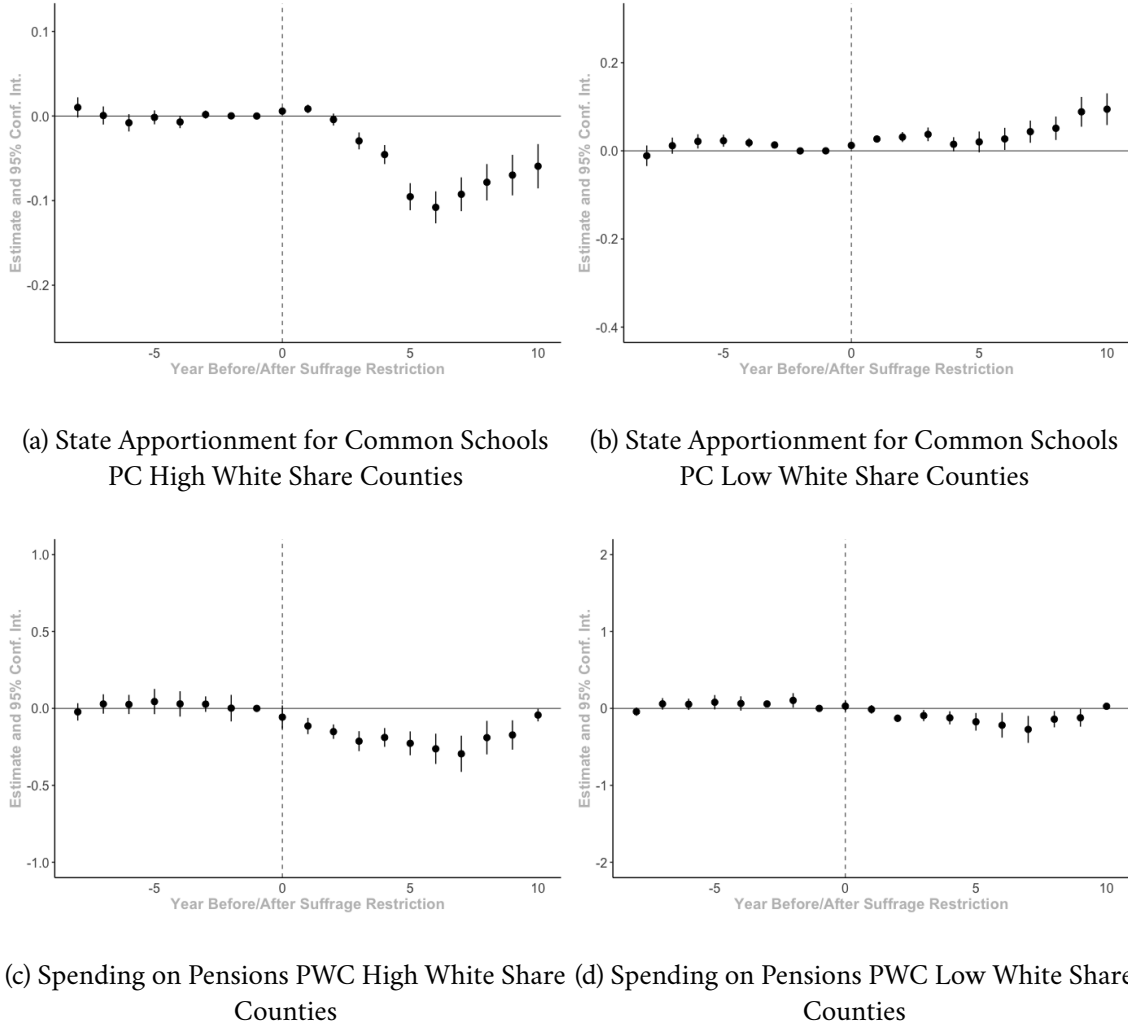
Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel A reports results for per capita state education funding by county; Panel B for pension spending per White capita. All models include county fixed effects and state-specific trends. Columns (2) and (4) add time-invariant controls—county output, population, urban share, and property value per White capita in 1880—interacted with linear time trends. Columns (3) and (4) include an interaction between suffrage restrictions and the 1880 local White population share.

Figure A.5: Marginal Effects: The Moderating Role of White Population Share on the Relationship Between Disenfranchisement and State Spending on Pensions per White Capita



Notes: Marginal effects of disenfranchisement on spending on pensions per white capita across counties by White population share. The figure is based on results from model (4) in Table 3, which accounts for year and county fixed effects, along with state-specific linear time trends, and the following pre-treatment controls interacted with year indicators: county output, population size, the share of urban population, and the assessed value of property per White capita in 1880.

Figure A.6 Event-Study Analysis of Suffrage Restrictions on Education and Pension Spending: Variation by Counties' White Population Share



Notes: Sun and Abraham (2021) approach for estimating dynamic treatment effects in event studies with heterogeneous treatment effects. Panels (a) and (b) display the effects on per capita school apportionment for high- and low-White-share counties, respectively, while panels (c) and (d) show the effects on pension spending per White capita for the same county groups. The models include county output per capita, urban population percentage, and total population (in logs), all measured in 1880, each interacted with year indicators to address potential violations of parallel trends in TWFE. The school spending models use the full sample of counties in 11 disenfranchising states, while the pension analysis is limited to counties in 7 states (Alabama, Virginia, Florida, Mississippi, South Carolina, and North Carolina) for which pension data are available. Standard errors are clustered at the county level.

Table A.6: **States with Grandfather Clause: Heterogeneous Effects of Disenfranchisement**

Panel A: State Apportionment for Education per Capita		
	Education Spending per Capita	
	(1)	(2)
Suffrage Restriction	−0.006 (0.012)	0.003 (0.014)
Suffrage Restriction × White Population 1880 (%)	−0.081*** (0.018)	−0.059** (0.023)
Observations	11,453	11,127
R ²	0.849	0.920
County FE	✓	✓
Year FE	✓	✓
Time-invariant covariates		✓
State-specific trends		✓

Panel B: Pension Spending per White Capita		
	Pension Spending per White Capita	
	(1)	(2)
Suffrage Restriction	0.033** (0.015)	0.060** (0.025)
Suffrage Restriction × White Population 1880 (%)	−0.042* (0.024)	−0.171*** (0.039)
Observations	5,784	5,681
R ²	0.780	0.889
County FE	✓	✓
Year FE	✓	✓
Time-invariant covariates		✓
State-specific trends		✓

Notes: All models include year and county fixed effects, and a dummy for the adoption of grandfather clauses (equal to 1 post-implementation) interacted with White population share. The analysis is limited to counties in states that adopted such clauses. Column (2) includes state-specific linear trends and pre-treatment controls—county output, population size, urban share, and assessed property value per White capita in 1880—interacted with year indicators. * p<0.10, ** p<0.05, *** p<0.01.

Table A.7: **Accounting for Literacy and Third-Party Vote: Heterogeneous Effects of Disenfranchisement**

Panel A: State Apportionment for Education per Capita		
	Education Spending per Capita	
	(1)	(2)
Suffrage Restriction	0.009 (0.048)	0.036*** (0.014)
Suffrage Restriction $\times$ White Population 1880 (%)	-0.069*** (0.020)	-0.062*** (0.019)
Suffrage Restriction $\times$ Literate Males (log)	0.003 (0.007)	
Suffrage Restriction $\times$ Third Party Vote (log)		-0.004 (0.003)
Observations	22,286	22,286
R ²	0.959	0.959
County FE	✓	✓
Year FE	✓	✓
Time-invariant covariates	✓	✓
State-specific trends	✓	✓

Panel B: Pension Spending per White Capita		
	Pension Spending per White Capita	
	(1)	(2)
Suffrage Restriction	-0.272 (0.257)	-0.024 (0.041)
Suffrage Restriction $\times$ White Population 1880 (%)	-0.200*** (0.049)	-0.183*** (0.048)
Suffrage Restriction $\times$ Literate Males (log)	0.039 (0.036)	
Suffrage Restriction $\times$ Third Party Vote (log)		0.008* (0.005)
Observations	9,946	9,946
R ²	0.796	0.796
County FE	✓	✓
Year FE	✓	✓
Time-invariant covariates	✓	✓
State-specific trends	✓	✓

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models include year and county fixed effects, as well as state-specific linear trends. The analysis is restricted to counties in states that implemented disenfranchisement. The base term for the White population share in 1880 is absorbed by county fixed effects. Column (1) includes the number of literate males (log) in 1880; column (2) includes the third-party gubernatorial vote share from the last election before disenfranchisement. All controls are interacted with the suffrage restriction indicator. Additional controls include county output, population size, urban share, and property value per White capita in 1880, each interacted with year indicators. Literacy data: 1880 full-count census (Ruggles et al. 2021); third-party vote data: Amlani and Algara (2021).

Table A.8: **Predictors of Disenfranchisement Timing: Time-Invariant and Time-Varying Covariates**

Panel A: Time-Invariant Predictors								
	1890 Cohort (1)	1891 Cohort (2)	1893 Cohort (3)	1896 Cohort (4)	1900 Cohort (5)	1902 Cohort (6)	1907 Cohort (7)	All Treated (8)
Population (log)	−2.749 (3.036)	−0.398 (1.839)	1.225 (1.997)	1.623 (1.680)	5.463* (2.495)	−1.749 (2.065)	−2.272 (2.048)	1.143 (1.519)
Output (log)	1.456 (1.675)	0.384 (1.014)	−0.413 (1.101)	−1.157 (0.927)	−3.373** (1.376)	0.732 (1.139)	1.293 (1.130)	−1.078 (0.838)
White Pop. (%)	−0.816 (1.697)	0.897 (1.028)	−0.156 (1.116)	−0.226 (0.939)	2.440 (1.394)	−1.347 (1.154)	−1.645 (1.144)	−0.852 (0.849)
Urban Pop. (%)	−6.569 (7.064)	−1.303 (4.279)	2.970 (4.646)	4.802 (3.909)	10.263 (5.805)	−4.329 (4.805)	−5.509 (4.765)	0.324 (3.534)
Public Officials (log)	0.867 (1.549)	−0.306 (0.938)	−0.845 (1.019)	−0.335 (0.857)	−1.785 (1.273)	1.264 (1.054)	1.162 (1.045)	0.023 (0.775)
Observations	14	14	14	14	14	14	14	14
R ²	0.338	0.384	0.273	0.486	0.553	0.223	0.236	0.834

Panel B: Time-Varying Predictors								
	1890 Cohort (1)	1891 Cohort (2)	1893 Cohort (3)	1896 Cohort (4)	1900 Cohort (5)	1902 Cohort (6)	1907 Cohort (7)	All Cohorts (8)
Population (log)	−0.027 (0.110)	0.081 (0.090)	0.027 (0.041)	−0.051 (0.059)	0.002 (0.061)	−0.042 (0.047)	−0.076 (0.093)	−0.085 (0.188)
Output (log)	−0.004 (0.060)	−0.023 (0.039)	−0.028 (0.034)	0.054 (0.051)	0.044 (0.059)	−0.008 (0.015)	0.054 (0.069)	0.090 (0.116)
White Pop. (%)	0.596 (0.494)	0.241 (0.266)	−0.012 (0.096)	−0.093 (0.121)	0.093 (0.246)	0.420 (0.322)	−0.490 (0.517)	0.754 (0.742)
Urban Pop. (%)	0.291 (0.404)	−0.241 (0.307)	0.056 (0.064)	−0.255 (0.152)	−0.204 (0.455)	0.082 (0.186)	0.037 (0.135)	−0.234 (0.797)
Number of Lynchings	0.000 (0.002)	−0.002 (0.002)	0.004 (0.005)	0.009 (0.009)	−0.004 (0.007)	0.003 (0.003)	−0.001 (0.002)	0.009 (0.011)
Observations	338	338	338	338	338	338	338	338
R ²	0.294	0.122	0.122	0.137	0.293	0.127	0.124	0.168

Notes: Panel A reports OLS regressions predicting the year of suffrage restriction adoption using time-invariant covariates measured in 1880. Panel B reports two-way fixed effects regressions using interpolated time-varying covariates from the census. The dependent variable in each column is an indicator for treatment in a specific year or any treatment (column 8). Only treated states are included in Panel B.

Table A.9: Accounting for the Secret Ballot: Heterogeneous Effects of Disenfranchisement

Panel A: State Apportionment for Education per Capita		
	Education Spending per Capita	
	(1)	(2)
Suffrage Restriction	0.030*	0.041**
	(0.015)	(0.016)
Suffrage Restriction $\times$ White Population 1880 (%)	−0.084***	−0.094***
	(0.023)	(0.023)
Observations	20,439	18,245
R ²	0.939	0.963
County FE	✓	✓
Year FE	✓	✓
State-specific trends	✓	✓
Time-invariant covariates		✓

Panel B: Pension Spending per White Capita		
	Pension Spending per White Capita	
	(1)	(2)
Suffrage Restriction	0.254***	0.157***
	(0.064)	(0.059)
Suffrage Restriction $\times$ White Population 1880 (%)	−0.327***	−0.267***
	(0.093)	(0.095)
Observations	6,088	5,812
R ²	0.760	0.805
County FE	✓	✓
Year FE	✓	✓
State-specific trends	✓	✓
Time-invariant covariates		✓

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models include year and county fixed effects and state-specific linear time trends. Column (2) extends the baseline specification by adding county output, population size, urban population share, and assessed property value per White capita in 1880—all interacted with year indicators. The base term for the 1880 White population share is absorbed by county fixed effects. The analysis excludes states that adopted the secret ballot prior to disenfranchisement—namely, Virginia and Alabama. Note that Alabama was already absent from the original pension sample.

Table A.10: **Accounting for 8-Box Law: Heterogeneous Effects of Disenfranchisement (Excluding South Carolina)**

Panel A: State Apportionment for Education per Capita		
	Education Spending per Capita	
	(1)	(2)
Suffrage Restriction	0.022* (0.013)	0.025* (0.014)
Suffrage Restriction $\times$ White Population 1880 (%)	−0.072*** (0.020)	−0.065*** (0.020)
Observations	23,672	21,428
R ²	0.938	0.960
County FE	✓	✓
Year FE	✓	✓
State-specific trends	✓	✓
Time-invariant covariates		✓

Panel B: Pension Spending per White Capita		
	Pension Spending per White Capita	
	(1)	(2)
Suffrage Restriction	−0.049* (0.027)	−0.039 (0.046)
Suffrage Restriction $\times$ White Population 1880 (%)	−0.141*** (0.039)	−0.160*** (0.053)
Observations	9,581	9,253
R ²	0.770	0.796
County FE	✓	✓
Year FE	✓	✓
State-specific trends	✓	✓
Time-invariant covariates		✓

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models include year and county fixed effects and state-specific linear time trends. Column (2) adds county-level covariates—output, population size, urban share, and assessed property value per White capita in 1880—interacted with year indicators. The 1880 White population share is absorbed by county fixed effects. South Carolina is excluded because it implemented 8-box laws prior to disenfranchisement.

Table A.11: Accounting for Secret Ballot, 8-Box Laws, and Excluding Georgia and North Carolina: Heterogeneous Effects of Disenfranchisement

Panel A: State Apportionment for Education per Capita		
	Education Spending per Capita	
	(1)	(2)
Suffrage Restriction	0.096*** (0.017)	0.094*** (0.017)
Suffrage Restriction $\times$ White Population 1880 (%)	-0.113*** (0.025)	-0.081*** (0.024)
Observations	17,760	15,617
R ²	0.940	0.969
County FE	✓	✓
Year FE	✓	✓
State-specific trends	✓	✓
Time-invariant covariates		✓

Panel B: Pension Spending per White Capita		
	Pension Spending per White Capita	
	(1)	(2)
Suffrage Restriction	-0.012 (0.032)	-0.024 (0.046)
Suffrage Restriction $\times$ White Population 1880 (%)	-0.226*** (0.057)	-0.223*** (0.058)
Observations	7,983	7,706
R ²	0.768	0.795
County FE	✓	✓
Year FE	✓	✓
State-specific trends	✓	✓
Time-invariant covariates		✓

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models include year and county fixed effects, as well as state-specific linear trends. Georgia is excluded from both panels (and was already omitted from the original pension sample); South Carolina is excluded because of its adoption of the eight-box law; and North Carolina is excluded because of its 1889 law. All models also control for the adoption of the secret ballot using indicators interacted with the local White population share. Column (2) adds county-level covariates—output, population, urban share, and property value per White capita in 1880—interacted with year indicators. The 1880 White population share is absorbed by county fixed effects. The analysis includes only counties in states that enacted disenfranchisement.

Table A.12: **Variation by State Financing Structure: Heterogeneous Effects of Disenfranchisement on State Apportionment for Education Per Capita**

	State Apportionment for Education per Capita	
	(1)	(2)
Suffrage Restriction	−0.012 (0.012)	0.048*** (0.016)
Suffrage Restriction × White Population 1880 (%)	−0.005 (0.019)	−0.072*** (0.025)
Observations	6,314	15,972
R ²	0.933	0.953
County FE	✓	✓
Year FE	✓	✓
State-specific time trends	✓	✓
Time-invariant covariates	✓	✓

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.010$. The dependent variable is per capita state education funding by county. All models include year and county fixed effects and state-specific linear trends, and are limited to counties in states that implemented disenfranchisement. Column (1) covers states with low local-to-state school tax ratios (NC, SC, AL, MS, VA, GA, TX); Column (2) includes states more reliant on local financing (LA, TN, AR, FL), based on the closest pre-disenfranchisement year with available data. The 1880 White population share is absorbed by county fixed effects. Models also control for 1880 county output, population, urban share, and property value per White capita, each interacted with year indicators.

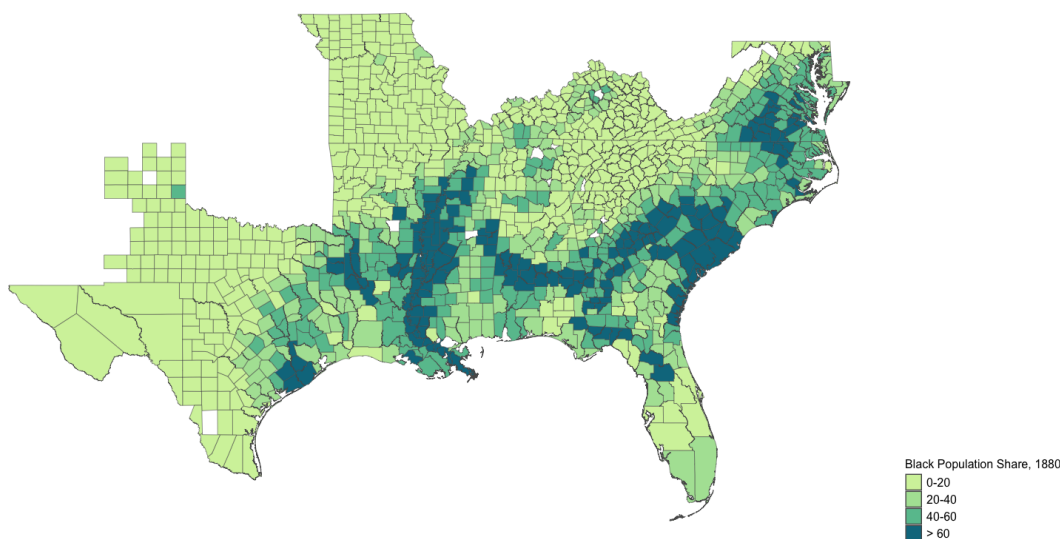
Appendix B: Black Disenfranchisement and Statewide White Political Control

In this appendix, we provide evidence for two claims made in the main text. First, we show that there was no bias in state legislative representation in favor of high Black-share counties prior to the disenfranchisement of Blacks. That is, representation was largely spatially neutral if total population was the basis of legislative apportionment. Second, we show that Black disenfranchisement also reduced the electoral need of Whites in low Black-share areas to ensure that Democratic gubernatorial candidates won election.

B.1 Legislative Apportionment Prior to Black disenfranchisement

The adoption of these disenfranchising measures, combined with a less frequently recognized institutional change introduced in the Reconstruction constitutions of the late 1860s—specifically, state legislative apportionment based largely on total population—increased the electoral weight of high Black-population share counties relative to the antebellum period. This happened because the Black population was highly concentrated in former plantation regions, a pattern closely mirroring the geography of slavery in 1860. Figure B.1 illustrates this spatial concentration, showing each county's Black population share in 1880, the initial year of our study. Once Black voters were disenfranchised, the electoral weight of the relatively few White voters in these counties was amplified substantially.

Figure B.1: **Black Population Share across Counties, 1880**



To demonstrate this we construct a measure of counties' relative representation in state legislatures in 1910 using the approach introduced by David and Eisenberg (1962) and Ansolabehere et al. (2002), which, like these authors, we refer to as the *Relative Representation Index* (RRI). The RRI captures whether a county is over- or under-represented in the state legislature relative to a given population baseline—either total population or White population. We use this index to demonstrate how the geographic distribution of White representation shifted after Black disenfranchisement. If disenfranchisement had a geographically neutral effect on White representation, the RRI would re-

main unchanged when shifting the baseline from total population to White population. To calculate this measure, we first determine each county's legislative representation by dividing the number of representatives it was apportioned in each legislative chamber by its total population. We then normalize this ratio by the corresponding state-level ratio of total representatives to total population, resulting in a county-specific measure of relative representation in each chamber. An RRI above 1 indicates that a county was overrepresented relative to "fair representation" based on a total population, while a value below 1 signals underrepresentation. As with Ansolabehere et al. (2002), our final measure of county-level political inequality is the logarithmic average of the RRI from both chambers of the bicameral state legislature.

To analyze how Black disenfranchisement influenced political representation, we construct three county-level versions of the RRI, each serving as a dependent variable in our analysis. Two of these measures use total population as the benchmark for fair representation—one calculated at the beginning (1880) and another at the end (1910) of our period—while the third measure, also calculated for 1910, uses only the White population as its benchmark. The two measures based on total population (1880 and 1910) allow us to confirm that formal legislative apportionment rules remained largely unchanged during this period. Meanwhile, the 1910 measure based exclusively on the White population enables us to assess whether the exclusion of Black voters resulted in the political overrepresentation of Whites in counties with large Black populations.

Because our goal is to assess whether there was a bias in White representation based on residential patterns, our main explanatory variable is each county's White population share. Given substantial demographic heterogeneity among Southern states, we present our results separately for three distinct state groupings, which we maintain throughout our analysis. First, we pool counties from the six "Deep South" states where Blacks represented at least 40% of the population: Alabama, Florida, Georgia, Louisiana, Mississippi, and South Carolina. Results for this group appear in Panel A of Table B.1, columns 1 (1880) and 2 (1910). Panel B includes counties from the three states with moderate Black populations (25-40%): Arkansas, North Carolina, and Virginia. Finally, Panel C contains counties from the two states with Black populations below 25%—Tennessee (21.7%) and Texas (17.7%). All models include state fixed effects, and standard errors are clustered at the state level.

Among the six high Black-share states (Panel A) and three mid Black-share states (Panel B), respectively, we observe a slight positive association, indicating marginal overrepresentation of counties with larger White population shares in both 1880 (column 1) and 1910 (column 2). This confirms that representation was not biased in favor of high Black-share counties when total population is used as the basis of relative representation either before or after Black disenfranchisement. Furthermore, in both sets of states, county White-population share captures little of the variation in the RRI. Surprisingly, the only states where representation showed a slight bias in favor of counties with higher Black population shares were Tennessee and Texas—the two with the lowest overall Black population proportions. This anomaly is likely due to the geographic concentration of Black populations in a few rural counties, a pattern not observed in any of the remaining nine higher Black-population share states.

While the relative representation based on total population remained largely unchanged, Black disenfranchisement dramatically reshaped the geographic distribution of White political influence. We can see this when we use the same apportionment data from 1910, but make White population—the only relevant voting group after disenfranchisement—the relevant basis for determining fair representation. If Black disenfranchisement had not altered the spatial distribution of White political power, we would expect no relationship between the RRI and a county’s White population share (as shown in columns 1 and 2, which use total population as the basis for representation). However, this pattern shifts dramatically in column 3 of Table B.1. While counties’ White population shares previously showed little correlation with their relative representation, we now observe a pronounced negative relationship: counties with higher White population shares became significantly underrepresented. Specifically, in the highest Black-share states, for every 1% increase in county White share, its relative representation in the state legislature falls by more than 2%. Furthermore, there is a substantial increase in R^2 when switching from total to White population, confirming that disenfranchisement fundamentally altered the political geography of legislative power.

Table B.1: **County Relative Representation (RRI) Using a Total Population Basis versus a White Population Basis, 1880 and 1910**

	RRI using Total Pop. 1880 (1)	RRI using Total Pop. 1910 (2)	RRI using White Pop. 1910 (3)
A) High Black-pop. Share State (>40%)			
White-pop. Share (%)	0.372** (0.123)	0.053 (0.176)	-2.117*** (0.297)
Counties	417	441	441
R ²	0.12	0.03	0.57
B) Mid Black-pop. Share State (25-40%)			
White-pop. Share (%)	0.092 (0.124)	0.115 (0.113)	-1.523*** (0.117)
Counties	272	289	289
R ²	0.03	0.02	0.6
C) Low Black-pop. Share State (<25%)			
White-pop. Share (%)	-1.101* (0.22)	-1.71* (0.549)	-3.098*** (0.494)
Counties	335	339	339
R ²	0.05	0.17	0.36

Notes: The dependent variable in each column is the county value for relative representation in the state legislature (*RRI*) in 1880 (column 1) and 1910 (columns 2 and 3). In columns 1 and 2, total population in 1880 (column 1) and 1910 (column 2) is used as the basis to determine the county's representation and "fair representation" for the state—along with the actual number of representatives apportioned to each county circa 1880 (column 1) and 1910 (column 2). In column 3, White population in 1910 is used to calculate *RRI*. Each model includes state fixed effects and standard errors clustered at the state level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Appendix C: Alternative Explanations

This section details the fiscal changes states adopted alongside voting restrictions; only one—Alabama—offers a possible alternative explanation for the expenditure shifts we observe. Because states rarely set floors or ceilings for prison, college, or pension spending, but occasionally did so for school taxes, our discussion focuses on common schools.

Of the fourteen states in our sample, four (Arkansas, Florida, Tennessee, and Texas) changed suffrage rules by statute, leaving school-spending mandates unchanged; North Carolina and Georgia used constitutional amendments with no fiscal provisions; and Alabama, Louisiana, Mississippi, South Carolina, and Virginia adopted new constitutions through conventions. These categories denote the principal *de jure* measure we code as treatment onset, not each state's full repertoire of sup-

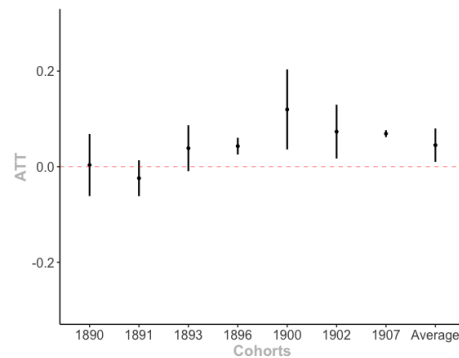
pression. As noted in the main text, several states—North Carolina among them (Edmonds 1951)—had earlier relied on statutory and administrative suppression that our coding treats as pre-treatment; accounting for these measures does not affect our results (Tables A.9–A.11).

Only Louisiana imposed a binding school-spending constraint: a minimum property-tax rate of 1.25 mills (of a 6-mill maximum). This cuts against our argument, implying that absent such floors school spending might have fallen further after suffrage restrictions. The constitution also capped state-university funding at \$15,000—repealed in 1904 in favor of language encouraging larger appropriations—and its cap on Confederate pensions was repeatedly replaced with more generous provisions, consistent with elites revising priorities after consolidating power. Louisiana’s concurrent fiscal rules thus ran against, not with, the shifts we observe, suggesting some actors anticipated the risks of suffrage restriction and moved preemptively to blunt them.

Other states made largely symbolic gestures. Mississippi mandated at least four months of free education but appropriated no specific school funds; South Carolina reaffirmed an existing 3-mill school tax and directed alcohol revenues to schools (though prior practice is unclear); and Virginia set a 1-to-5-mill school-tax range identical to its earlier provisions.

Alabama presents the only potential threat to inference. This is because changes in state finances could stem from other constitutional provisions adopted in tandem with suffrage restrictions in the state. Yet these changes run counter to our argument, thereby reducing the risk that they may be the primary drivers of our results. Specifically, Alabama established a maximum property tax rate of 6.5 mills and designated a mandatory 3-mill tax for schools, marking an increase from previous measures. Thus, these shifts suggest our findings might provide a conservative estimate of the potential impact of suffrage restrictions on school spending. Additionally, Alabama set a minimum level of funding for the University of Alabama. Although the state ultimately appropriated significantly more than this minimum (\$35,000), one might be concerned that this provision could underlie the increase in college expenditures we observe following disenfranchisement. To alleviate this concern, we replicate our college spending analyses excluding Alabama from the sample. The results, displayed in Figure C.1, show that our findings remain unchanged.

Figure C.1: **Overall Effect of Suffrage Restrictions on College Expenditures, excluding Alabama: Average and Cohort-Specific ATT Estimates**



Notes: Cohort-specific ATT estimates for each adoption year and average across years, using Callaway and Sant’Anna (2021)’s doubly robust method. The sample includes 13 Southern states (excluding Alabama), with three never-treated. Standard errors are clustered at the state level.

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